MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

## GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.

Each subject is divided into 2 sections:

**Section 1:** Single Correct Type Questions (4 options, only one correct)

**Section 2:** Numerical Value Type Questions (answer to be entered as numerical value)

Marking Scheme:

**Single Correct:** +4 for correct, -1 for wrong, 0 if unattempted.

**Numerical Value:** +4 for correct, -1 for wrong.

By: C I P H Ξ R

**Q.1**

In the Young's double slit experiment, the intensities at two points $P_1$ and $P_2$ on the screen are respectively $I_1$ and $I_2$. If $P_1$ is located at the centre of a bright fringe and $P_2$ is located at a distance equal to a quarter of fringe width from $P_1$, then $\frac{I_1}{I_2}$ is

**(A)** 2

**(B)** $\frac{1}{2}$

**(C)** 4

**(D)** 16

**Q.2**

$A, B, C$ are points in a vertical line such that $AB = BC$. If a body falls freely from rest at $A$, $t_1$ and $t_2$ are times taken to travel distances $AB$ and $BC$, then ratio $(t_2/t_1)$ is

**(A)** $\sqrt{2}+1$

**(B)** $\sqrt{2}-1$

**(C)** $2\sqrt{2}$

**(D)** $\frac{1}{\sqrt{2}-1}$

**Q.3**

Consider a concave mirror of $10\ \text{cm}$ focal length illuminated by an object kept at a distance of $25\ \text{cm}$. The distance at which the image is formed and its magnification respectively are

**(A)** $-16.7\ \text{cm}$ and -0.67

**(B)** $7.1\ \text{cm}$ and 0.29

**(C)** $-16.7\ \text{cm}$ and 0.67

**(D)** $7.1\ \text{cm}$ and -0.29

**Q.4**

If the pressure of a gas increases by $2\%$ at constant volume, then its temperature

**(A)** increases by $2\%$

**(B)** decreases by $2\%$

**(C)** does not change

**(D)** decreases by $1\%$

**Q.5**

The length of an elastic string is $a$ metres when the longitudinal tension is $4\ \mathrm{N}$ and $b$ metres when the longitudinal tension is $5\ \mathrm{N}$. The length of the string in metres when the longitudinal tension is $9\ \mathrm{N}$ is

**(A)** $a - b$

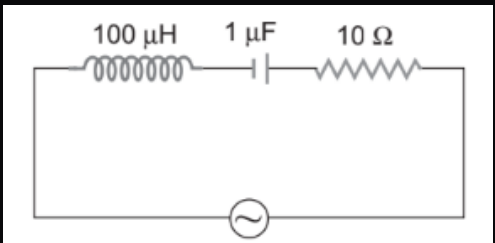
**(B)** $5b - 4a$

**(C)** $2b - \frac{1}{4}a$

**(D)** $4a - 3b$

**Q.6**

The following series $L - C - R$ circuit, when driven by an emf source of angular frequency 70 kilo-radians per second, the circuit effectively behaves like



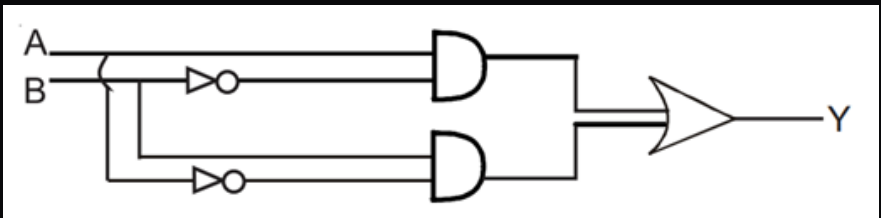

**(A)** purely resistive circuit

**(B)** series R-L circuit

**(C)** series R-C circuit

**(D)** series L-C circuit with R = 0

**Q.7**




For the given logic circuit, the output $Y$ for inputs $(A = 0, B = 1)$ and $(A = 0, B = 0)$ respectively are

**(A)** 0,0

**(B)** 0,1

**(C)** 1,0

**(D)** 1,1

**Q.8**

A physical quantity $P$ is related to four observables $a, b, c$ and $d$ as $P = \frac{\sqrt{ab} \cdot d^{\alpha}}{\sqrt{c}}$ ($\alpha$ is constant) The percentage errors in $a, b, c$ and $d$ are $0.5\%$ in each. If the percentage error in $P$ is $2\%$, then $\alpha$ is

**(A)** $\frac{5}{2}$

**(B)** $\frac{2}{5}$

**(C)** $\frac{3}{4}$

**(D)** $\frac{3}{2}$

**Q.9**

Three closed vessels $A, B$ and $C$ are at the same temperature $T$ and contain gases. Vessel $A$ contains only $O_2$, $B$ contains only $N_2$ and C contains a mixture of equal quantities of $O_2$ and $N_2$. If the rms speed of $O_2$ molecules in vessel $A$ is $v_1$ and that of $N_2$ molecules in vessel $B$ is $v_2$ then the rms speed of $O_2$ molecules in vessel C is

**(A)** $\left(\frac{v_1+v_2}{2}\right)$

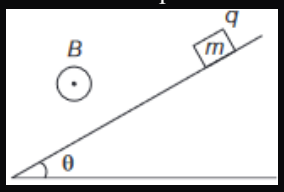
**(B)** $v_1$

**(C)** $(v_1 v_2)$

**(D)** $\frac{v_1}{2}$

**Q.10**

A small block of mass $20\ \text{g}$ and charge $4\text{mC}$ is released on a long smooth inclined plane of inclination angle of $45^\circ$, A uniform horizontal magnetic field of $1\ \text{T}$ is acting parallel to the surface, as shown in the figure. The time from the start when the block loses contact with the surface of the plane is




**(A)** $2\ \text{s}$

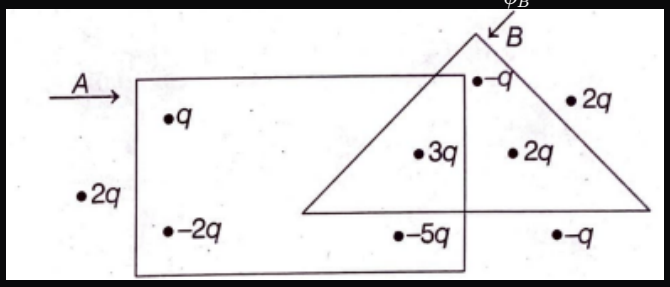
**(B)** $3\ \text{s}$

**(C)** $5\ \text{s}$

**(D)** $6\ \text{s}$

**Q.11**

The distribution of some charges on two Gaussian surfaces $A$ and $B$ are as shown in the figure. If $\phi_A$ and $\phi_B$ are electric fluxes linked with the surfaces $A$ and $B$ respectively. then $\frac{\phi_A}{\phi_B} =$




**(A)** $-\frac{1}{5}$

**(B)** $-3$

**(C)** $-\frac{3}{2}$

**(D)** $-\frac{3}{4}$

**Q.12**

A spherical body of density $\rho$ is floating half immersed in a liquid of density $d$. If $\sigma$ is the surface tension of the liquid, then the radius of the body is (Assuming contact angle 0)

**(A)** $\sqrt{\frac{3\sigma}{g(d-2\rho)}}$

**(B)** $\sqrt{\frac{6\sigma}{g(2\rho-d)}}$

**(C)** $\sqrt{\frac{4\sigma}{g(2\rho-d)}}$

**(D)** $\sqrt{\frac{12\sigma}{g(2\rho-d)}}$

**Q.13**

A body of mass $4\ \text{kg}$ is moving with momentum of $8\ \text{kg ms}^{-1}$. A force of $0.2\ \text{N}$ acts on it in the direction of motion of the body for 10 seconds. The increase in kinetic energy in joules is

**(A)** $10$

**(B)** $8.5$

**(C)** $4.5$

**(D)** $4$

**Q.14**

For photoelectric effect which of the following statements are true.
I) The kinetic energies of the photoelectrons do not depend on the frequency of light
II) Photoelectric effect will always occur for highly intense light
III) The maximum kinetic energy of photoelectron does not depend upon the intensity of the light
IV) The escaping electron's kinetic energy is larger for larger frequency

**(A)** I and II only

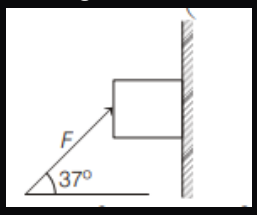
**(B)** II and III only

**(C)** III and IV only

**(D)** IV and I only

**Q.15**

A block of mass $2\ \text{kg}$ is being pushed against a wall by a force $F = 90\ \text{N}$ as shown in the figure. If the coefficient of friction is 0.25 , then the magnitude of acceleration of the block is (Take, $g = 10\ \text{ms}^{-2}$) $\left(\sin 37^\circ = \frac{3}{5}\right)$




**(A)** $16\ \text{ms}^{-2}$

**(B)** $8\ \text{ms}^{-2}$

**(C)** $38\ \text{ms}^{-2}$

**(D)** $54\ \text{ms}^{-2}$

**Q.16**

In a hypothetical Bohr hydrogen atom, if the mass of the electron is doubled then the energy of the electron in the first orbit is

**(A)** $-27.2\text{eV}$

**(B)** $-13.6\text{eV}$

**(C)** $-6.8\text{eV}$

**(D)** $-3.4\text{eV}$

**Q.17**

A nichrome wire $50\ \text{cm}$ long and $1\ \text{mm}^2$ cross-section carries a current of $4\ \text{A}$. When connected to 2 volt battery. The resistivity of nichrome wire in $\Omega - \text{m}$ is

**(A)** $1 \times 10^{-6}$

**(B)** $4 \times 10^{-7}$

**(C)** $3 \times 10^{-7}$

**(D)** $2 \times 10^{-7}$

**Q.18**

Pick out the longest wavelength from the following types of radiations.

**(A)** Visible blue light

**(B)** Ultraviolet light

**(C)** X-rays

**(D)** Visible red light

**Q.19**

The radius of Earth is about $6400\ \text{km}$ and that of Mars is $3200\ \text{km}$, and mass of the Earth is about 10 times mass of Mars. An object weight $200\ \text{N}$ on the surface of Earth. Then, its weight on the surface of Mars will be

**(A)** $80\ \text{N}$

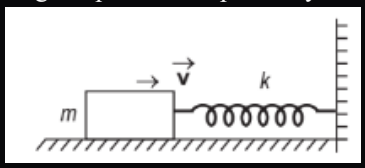
**(B)** $40\ \text{N}$

**(C)** $20\ \text{N}$

**(D)** $8\ \text{N}$

**Q.20**

A block of mass $m = 25\ \text{kg}$ sliding on a smooth horizontal surface with a velocity $v = 3\ \text{ms}^{-1}$ meets the spring of spring constant $k = 100\ \text{N/m}$ fixed at one end as shown in figure. The maximum compression of the spring and velocity of block as it returns to the original position respectively are




**(A)** $1.5\ \text{m}, -3\ \text{ms}^{-1}$

**(B)** $1.5\ \text{m}, 0.01\ \text{ms}^{-1}$

**(C)** $1.0\ \text{m}, 3\ \text{ms}^{-1}$

**(D)** $0.5\ \text{m}, 2\ \text{ms}^{-1}$

**Q.21**

If $B_1$ is the magnetic field induction at a point on the axis of a circular coil of radius $R$ situated at a distance $R\sqrt{3}$ and $B_2$ is the magnetic field at the centre of the coil, then the ratio of $\frac{B_1}{B_2}$ is equal to K, find 16K.

**Q.22**

The Young's modulus of a material is $2 \times 10^{11}\ \mathrm{N/m^2}$ and its elastic limit is $1 \times 10^{8}\ \mathrm{N/m^2}$. For a wire of $1\ \mathrm{m}$ length of this material, the maximum elongation achievable is K mm, find 6K (in mm).

**Q.23**

The velocity of an object moving in a straight line path is given as a function of time by $v = 6t - 3t^2$, where $v$ is in $\mathrm{ms^{-1}}$, $t$ is in $\mathrm{s}$. The average velocity of the object between, $t = 0$ and $t = 2\ \mathrm{s}$ is

**Q.24**

A coil has 1000 turns and $500\ \mathrm{cm^2}$ as its area. The plane of the coil is placed at right angles to a magnetic induction field of $2 \times 10^{-5}\ \mathrm{Wb/m^2}$. The coil is rotated through $180^\circ$ in $0.2\ \mathrm{s}$. The average emf induced in the coil, in $\mathrm{mV}$, is

**Q.25**

A particle of mass $15\ \mathrm{kg}$ is moving with a uniform speed $8\ \mathrm{ms^{-1}}$ in $xy$-plane along the line $3y = 4x + 10$, then the magnitude of its angular momentum about the origin in
$\mathrm{kg \cdot m^2\ s^{-1}}$ is $\ldots \left(\sin 53^\circ = \frac{4}{5}\right)$

**Q.26**

The IUPAC name of the following complex is $\left[\mathrm{Cr\left(NH_3\right)_3\left(H_2O\right)_2\ Cl}\right]\mathrm{Cl_2}$.

**(A)** Triamminediaqua chlorido chromium (III) ch!oride.

**(B)** Diaquatriammine chlorido chromium (III) chloride.

**(C)** Chlorido diaquatriammine chromium (III) chloride.

**(D)** Triammine diaqua trichlorido chromium (III).

**Q.27**

The order of strengths of the following carboxylic acids is
(i) $\mathrm{CH_3 - CH_2 - COOH}$
(ii) $\mathrm{CH_3 - COOH}$
(iii) $\mathrm{C_6H_5 - COOH}$
(iv) $\mathrm{C_6H_5 - CH_2COOH}$

**(A)** (iii) > (iv) > (ii) > (i)

**(B)** (iv) > (ii) > (iii) > (i)

**(C)** (iii) > (ii) > (iv) > (i)

**(D)** (i) > (iv) > (ii) > (iii)

**Q.28**

The stability of +1 oxidation state increases in the sequence

**(A)** $\mathrm{Ga < In < Al < TI}$

**(B)** Al < Ga < In < TI

**(C)** Tl < In < Ga < Al

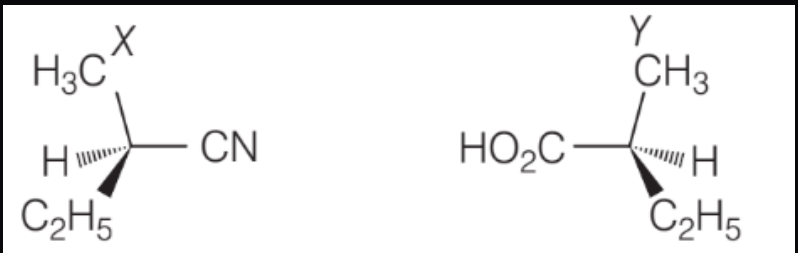
**(D)** In < Tl < Ga < Al

**Q.29**

What are $X$ and $Y$ in the following reaction sequence?




**(A)**



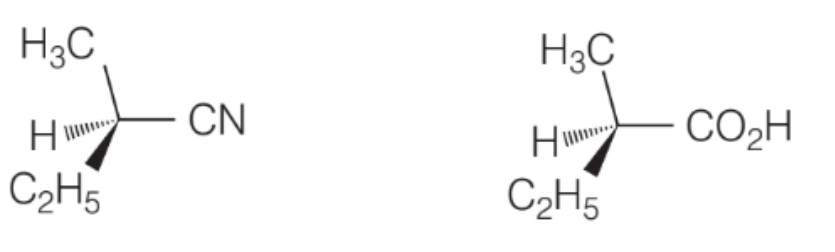

**(B)**




**(C)**




**(D)**




**Q.30**

An organic compound ' $X$ ' on treatment with pyridinium chloro chromate in dichloromethane gives compound ' $Y$ '. Compound ' $Y$ ', reacts with $I_2$ and alkali to form triiodomethane. The compound ' $X$ ' is

**(A)** $C_2H_5OH$

**(B)** $CH_3CHO$

**(C)** $CH_3COCH_3$

**(D)** $CH_3COOH$

**Q.31**

Identify the correct statements from the following :

I. The ionic radius of $Pr^{3+}$, $Dy^{3+}$ and $Sm^{3+}$ follow the order, $Sm^{3+} > Pr^{3+} > Dy^{3+}$.

II. $Eu^{2+}$ acts as strong reducing reagent.

III. Pu exhibits +7 oxidation state.

**(A)** I, II only

**(B)** I, III only

**(C)** I, II, III

**(D)** II, III only

**Q.32**

Which of the following chemicals may be used to identify three unlabelled beakers containing conc.NaOH, conc.$H_2SO_4$ and water?

**(A)** $NH_4NO_3$

**(B)** $NaCl$

**(C)** HCOONa

**(D)** None of these

**Q.33**

The correct statement about $Cr^{2+}$ and $Mn^{3+}$ among the following is (Given, atomic numbers of $Cr = 24$ and $Mn = 25$ )

**(A)** $Cr^{2+}$ is a reducing agent

**(B)** $Mn^{3+}$ is a reducing agent

**(C)** Both $Cr^{2+}$ and $Mn^{3+}$ exhibit $d^5$ outer electronic configuration

**(D)** When $Cr^{2+}$ is used as reducing agent, it attains $d^5$ configuration

**Q.34**

If general formula of oxime and semicarbazone is $C = N - Z$, what is ' $Z$ ' in oxime (I) semicarbazone (II)?

**(A)** $NHCONH_2 \quad OH$

**(B)** $OH \quad NH_2$

**(C)** $OH \quad NHCONH_2$

**(D)** $NH_2OH$

**Q.35**

A compound absorbs light in the wavelength region $490 - 500\ nm$. Its complementary colour is

**(A)** red

**(B)** blue

**(C)** orange

**(D)** blue-green

**Q.36**

Identify the correct statements from the following
(i) Electromeric effect is a permanent effect.
(ii) Hyper conjugation is a temporary effect.
(iii) Fractional distillation is used to separate two liquids from a mixture if the difference in their boiling points is less.
(iv) Different compounds are adsorbed on an adsorbent different extents.

**(A)** ii, iii, iv

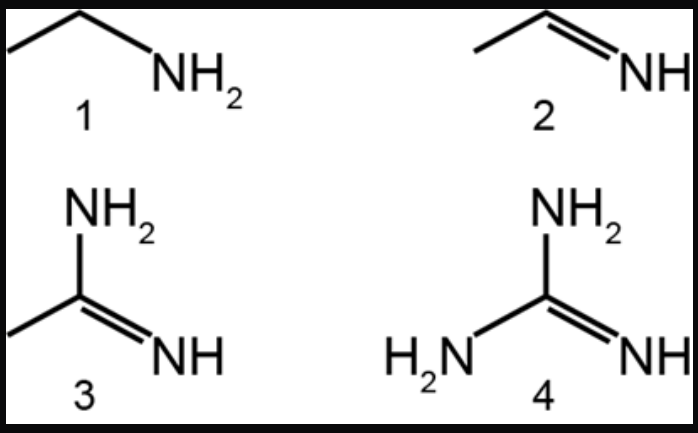
**(B)** i, ii, iii

**(C)** ii, iv

**(D)** iii, iv

**Q.37**

The correct order of basicity of the following compounds is




**(A)** $1 < 2 < 3 < 4$

**(B)** $1 < 2 < 4 < 3$

**(C)** $2 < 1 < 3 < 4$

**(D)** $4 < 3 < 2 < 1$

**Q.38**

The electronic transitions from $\mathrm{n} = 2$ to $\mathrm{n} = 1$ will produce shortest wavelength in (where $\mathrm{n} =$ principal quantum state)

**(A)** $\mathrm{Li}^{+2}$

**(B)** $\mathrm{He}^{+}$

**(C)** $\mathrm{H}$

**(D)** $\mathrm{H}^{+}$

**Q.39**

Which of the following are reducing sugars?
(A) Sucrose, (B) Maltose, (C) Lactose, (D) Fructose

**(A)** $A, B, C$

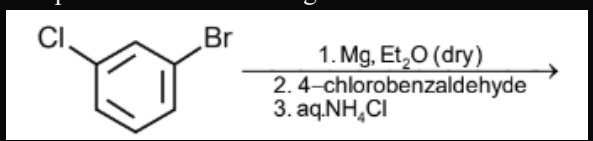
**(B)** $A, B, D$

**(C)** $A, C, D$

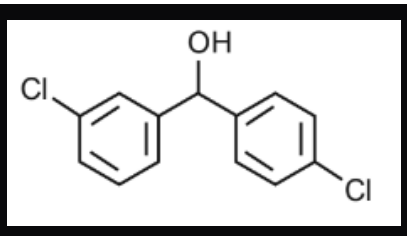
**(D)** $B, C, D$

**Q.40**

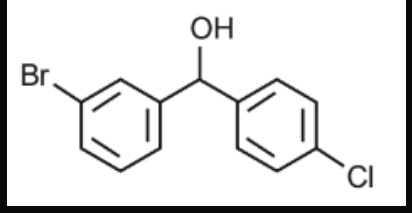
The product of the following reaction is :



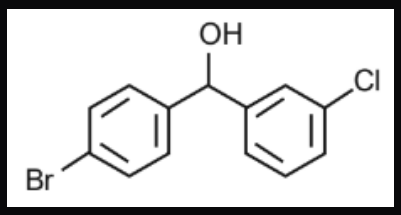

**(A)**



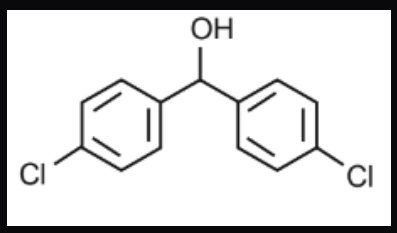
**(B)**




**(C)**




**(D)**




**Q.41**

Match the following

| List - I (Molecules) | List - II (Dipolemoment $\mu$, D) |
|---|---|
| A. $H_2O$ | I. 0 |
| B. $BF_3$ | II. 0.23 |
| C. $NH_3$ | III. 1.47 |
| D. $NF_3$ | IV. 1.85 |

The correct answer is

**(A)** $A-IV, B-I, C-II, D-III$

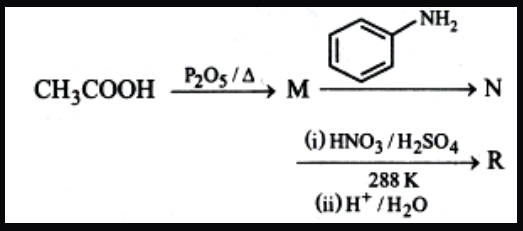
**(B)** A - IV, B - I, C - III, D - II

**(C)** A - IV, B - III, C - I, D - II

**(D)** A - III, B -IV, C - II, D - I

**Q.42**

What is the major product ' $R$ ' in the following reaction sequence?




**(A)** o-Nitroaniline

**(B)** m-Nitroaniline

**(C)** p-Nitroaniline

**(D)** p-Aminobenzene sulphonic acid

**Q.43**

In the modern periodic table, the acidic nature of oxides

**(A)** increases from left to right and decreases from top to bottom

**(B)** decreases from right to left and increases from top to bottom

**(C)** remains same across a period and increases from top to bottom

**(D)** decreases from left to right and remains the same down the group

**Q.44**

Assertion (A) $-\mathrm{NH_2}$ group of aniline is ortho, para directing in electrophilic substitutions.
Reason (R) $-\mathrm{NH_2}$ group stabilises the arenium ion formed by the ortho, para attack of the electrophile.
The correct answer is

**(A)** Both $(A)$ and $(R)$ are correct, $(R)$ is the correct explanation of $(A)$

**(B)** Both $(\mathrm{A})$ and $(\mathrm{R})$ are correct, $(\mathrm{R})$ is not the correct explanation of $(\mathrm{A})$

**(C)** (A) is correct, but (R) is not correct

**(D)** (A) is not correct, but (R) is correct

**Q.45**

The electronic configuration of elements $A$, $B$ and $C$ are $[\mathrm{He}]2s^1$, $[\mathrm{Ne}]3s^1$ and [Ar] $4s^1$ respectively, which one of the following order is correct for the first ionization potentials (in $\mathrm{kJmol^{-1}}$ ) of $A$, $B$ and $C$ ?

**(A)** $A > B > C$

**(B)** $C > B > A$

**(C)** $B > C > A$

**(D)** $C > A > B$

**Q.46**

$0.126\mathrm{g}$ of an acid is needed to completely neutralise $20\mathrm{mL}0.1$ (N) NaOH solution. The equivalent weight of the acid is

**Q.47**

Consider the following salts: $\mathrm{NaCl}, \mathrm{HgCl_2}, \mathrm{Hg_2Cl_2}, \mathrm{CuCl_2}, \mathrm{CuCl}$ and $\mathrm{AgCl}$. How many of these salts are insoluble in water?

**Q.48**

For a reaction $A + B \longrightarrow P$, the following data are provided

| Entry | [A] in M | [B] in M | Initial rate (M/s) |
|---|---|---|---|
| 1 | 0.02 | 0.02 | $2 \times 10^{-2}$ |
| 2 | 0.02 | 0.04 | $4 \times 10^{-2}$ |
| 3 | 0.04 | 0.04 | $8 \times 10^{-2}$ |

The rate constant for this reaction in standard unit is

**Q.49**

The atomisation enthalpy of $\mathrm{CH_4}$ is $1660\ \mathrm{kJ\ mol^{-1}}$. The $\mathrm{C-H}$ bond enthalpy of each successive step in $\mathrm{CH_4} \rightarrow \mathrm{CH_3} \rightarrow \mathrm{CH_2} \rightarrow \mathrm{CH}$ are $+15, +30$ and $+45\ \mathrm{kJ\ mol^{-1}}$ higher than the mean bond enthalpy of $\mathrm{CH}$ bonds, respectively. The bond enthalpy of the last $\mathrm{C-H}$ unit is (in KJ/mol)

**Q.50**

How many ions of the following have a bond order of $2.5$ ?
$N_2^-; NO^-; C_2^-; N_2^+; C_2^{2-}; CN^+$

**Q.51**

Let $a, b$ and $c$ be non-negative real numbers satisfying $a+b+c=9$. If the maximum value of the expression $a^2b^3c^4$ can be expressed as $2^x3^y$, where $x$ and $y$ are natural numbers, then the value of $\log_{10}\left(x^y\right)$, is:

**(A)** 2

**(B)** 3

**(C)** 4

**(D)** 6

**Q.52**

Find the sum of all possible integral value(s) of 'a' for which $F(x)=\frac{x^3}{3}+(a-3)x^2+x-13$ has negative point of local minimum in the interval $[1,100]$.

**(A)** 5040

**(B)** 5050

**(C)** 4990

**(D)** 4950

**Q.53**

Let $A(5,12), B(-13\cos\theta, 13\sin\theta)$ and $C(13\sin\theta, -13\cos\theta)$ are angular points of $\triangle ABC$ where $\theta \in R$. The locus of orthocentre of $\triangle ABC$ is

**(A)** $x-y+7=0$

**(B)** $x-y-7=0$

**(C)** $x+y-7=0$

**(D)** $x+y+7=0$

**Q.54**

Let the function $g: R \to \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ be given by $g(t)=\frac{\pi}{2}-2\cot^{-1}\left(3^{-t}\right)$. Then $g$ is

**(A)** Even and is strictly increasing in $(-\infty, \infty)$

**(B)** Odd and is strictly decreasing in $(-\infty, \infty)$

**(C)** Even and is strictly decreasing in $(-\infty, \infty)$

**(D)** Odd and is strictly increasing in $(-\infty, \infty)$

**Q.55**

If $\int \frac{x^2}{x^4+3x^2+9}dx = A\tan^{-1}\left(\frac{x^2-3}{3x}\right)+\frac{B}{\sqrt{3}}\ln\left|\frac{x^2-\sqrt{3}x+3}{x^2+\sqrt{3}x+3}\right|+C$, then find the value of $12(A+B)$

**(A)** 7

**(B)** 10

**(C)** 5

**(D)** 14

**Q.56**

The mean and the standard deviation of a data of 8 items are 25 and 5 respectively. If two items 15 and 25 are added to this data, then the variance of the new data is

**(A)** 29

**(B)** 24

**(C)** 26

**(D)** $\sqrt{29}$

**Q.57**

For some real number $\lambda$, if the area of the triangle having $\vec{a}=3\hat{i}-\hat{j}+\lambda\hat{k}$ and $\vec{b}=\lambda\hat{i}+\hat{j}-3\hat{k}$ as two of its sides is $\frac{\sqrt{195}}{2}$, then the number of distinct possible values of $\lambda$ is

**(A)** 4

**(B)** 3

**(C)** 2

**(D)** 1

**Q.58**

If $x^2+xy+y^2=k$, then $\frac{d^2y}{dx^2}=$

**(A)** $\frac{-6k}{(x+2y)^3}$

**(B)** $\frac{-6k}{x+2y^2}$

**(C)** $\frac{x^2+xy+y^2}{(2x+y)^2}$

**(D)** $0$

**Q.59**

If $z$ is a complex number satisfying the equation $|z-(1+i)|^2=2$ and $\omega=\frac{2}{z}$, then the locus traced by ' $\omega$ ' in the complex plane is

**(A)** $x-y-1=0$

**(B)** $x+y-1=0$

**(C)** $x-y+1=0$

**(D)** $x+y+1=0$

**Q.60**

For $a, b>0$, let $5a-b, 2a+b, a+2b$ be in A.P. and $(b+1)^2, ab+1, (a-1)^2$ are in G.P., then compute $\left(a^{-1}+b^{-1}\right)$

**(A)** 7

**(B)** 6

**(C)** 5

**(D)** 2

**Q.61**

3 women and 15 men are to be arranged in a row such that there should be atleast 2 men between all the two consecutive women. Then number of such arrangements is :

**(A)** ${}^{14}C_4 3$ !

**(B)** 3! 15!

**(C)** ${}^{14}C_{12}$

**(D)** ${}^{14}C_3$ 3! 15!

**Q.62**

Let $l_1$ be the line $4x + 3y = 3$ and $I_2$ be the line $y = 8x$. $L_1$ is the line formed by reflecting $l_1$ across the line $y = x$ and $L_2$ is the line formed by reflecting $I_2$ across the $x$-axis. If $\theta$ is the acute angle between $L_1$ and $L_2$ such that $\tan\theta = \frac{a}{b}$, where $a$ and $b$ are coprime then find $(a + b)$

**(A)** 71

**(B)** 63

**(C)** 57

**(D)** 27

**Q.63**

$A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$ then let us define a function $f(x) = \det\left(A^{\top}A^{-1}\right)$ then which of the following can not be the value of $\underbrace{f(f(f(f\ldots\ldots\ldots.....f(x))))}_{n \text{ times}}$ is $(n \geq 2)$

**(A)** $f^n(x)$

**(B)** 1

**(C)** $f^{\mathrm{n}-1}(\mathrm{x})$

**(D)** $nf(x)$

**Q.64**

Find the value of $32 \sin\frac{90^\circ}{17} \cos\frac{180^\circ}{17} \cos\frac{360^\circ}{17} \cos\frac{720^\circ}{17}$.

**(A)** 1

**(B)** 8

**(C)** 4

**(D)** 2

**Q.65**

Let $R$ be a relation defined on the set $Z$ of all integers and $xRy$, when $x + 2y$ is divisible by 3 , then

**(A)** $R$ is not transitive

**(B)** $A$ is symmetric only

**(C)** $R$ is an equivalence relation

**(D)** $A$ is not an equivalence relation

**Q.66**

The reflection of the point $P(1, 0, 0)$ in the line $\frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+10}{8}$ is

**(A)** $(3, -4, -2)$

**(B)** $(5, -8, -4)$

**(C)** $(1,-1,-10)$

**(D)** $(2,-3,8)$

**Q.67**

A function $y=f(x)$ satisfies $(x+1)\cdot f'(x)-2\left(x^2+x\right)f(x)=\frac{e^{x^2}}{(x+1)}, \forall x>-1$ If $f(0)=5$, then $f(x)$ is

**(A)** $\left(\frac{3x+5}{x+1}\right)\cdot e^{x^2}$

**(B)** $\left(\frac{6x+5}{x+1}\right)\cdot e^{x^2}$

**(C)** $\left(\frac{6\mathrm{x}+5}{(\mathrm{x}+1)^2}\right)\cdot \mathrm{e}^{\mathrm{x}^2}$

**(D)** $\left(\frac{5-6x}{x+1}\right)\cdot \mathrm{e}^{x^2}$

**Q.68**

Let a red die, a blue die, a green die and a white die are rolled once, the dice being fair. The out comes on the red, blue, green and white die denote the numbers $a, b, c$ and $d$ respectively. Let $E$ denotes the event, that absolute value of $(a-1)(b-2)(c-3)(d-6)=1$, then $P(E)$ is :

**(A)** $\frac{1}{324}$

**(B)** $\frac{1}{648}$

**(C)** $\frac{2}{324}$

**(D)** $\frac{1}{160}$

**Q.69**

If $\alpha=\sin\left(\cot^{-1}\left(\tan\left(\cos^{-1}\frac{2}{3}\right)\right)\right)$ and $\beta=\sin\left(\mathrm{cosec}^{-1}\left(\cot\left(\tan^{-1}\frac{1}{3}\right)\right)\right)$ are the roots of the quadratic equation $ax^2+bx+c=0$ where $a, b, c$ are integers and $c$ is prime then the value of $(a+b+c)$ equals

**(A)** 20

**(B)** 11

**(C)** 7

**(D)** 2

**Q.70**

Let $a, b, c \in R_0$ and distinct such that $\vec{\alpha}=a\hat{i}+b\hat{j}+c\hat{k}$ and $\bar{\beta}=(b-c)\bar{i}+(c-a)j+(a-b)k$ are parallel then $\frac{|\bar{\alpha}|}{|\bar{\beta}|}$ is -

**(A)** $\sqrt{3}$

**(B)** $\frac{1}{\sqrt{3}}$

**(C)** 2

**(D)** $\frac{1}{2}$

**Q.71**

If $n \in N$, then the highest integer $m$ such that $2^m$ divides $3^{2n+2}-8n-9$ is

**Q.72**

For $\mathrm{n} \in \mathrm{N}$, let $\mathrm{x_n}$ be defined as $\left(1+\frac{1}{\mathrm{n}}\right)^{\mathrm{n}+\mathrm{x_n}}=\mathrm{e}$, then $\mathrm{Lim_{n\to\infty}\, x_n}$ equals $K$, find $4K$

**Q.73**

The value of $\int_0^{100}\{\sqrt{x}\}dx$ (where $\{x\}$ is the fractional part of $x$ is (mark answer to nearest integer)

**Q.74**

In $\triangle ABC$, a point $P$ is chosen on side $\overrightarrow{AB}$ so that $AP:PB=1:4$ and a point $Q$ is chosen on the side $\overrightarrow{BC}$ so that $CQ:QB=1:3$. Segment $\overrightarrow{CP}$ and $\overrightarrow{AQ}$ intersect at $M$. If the ratio $\frac{MC}{PC}$ is expressed as a rational number in the lowest term as $\frac{a}{b}$, then find $(a+b)$.

**Q.75**

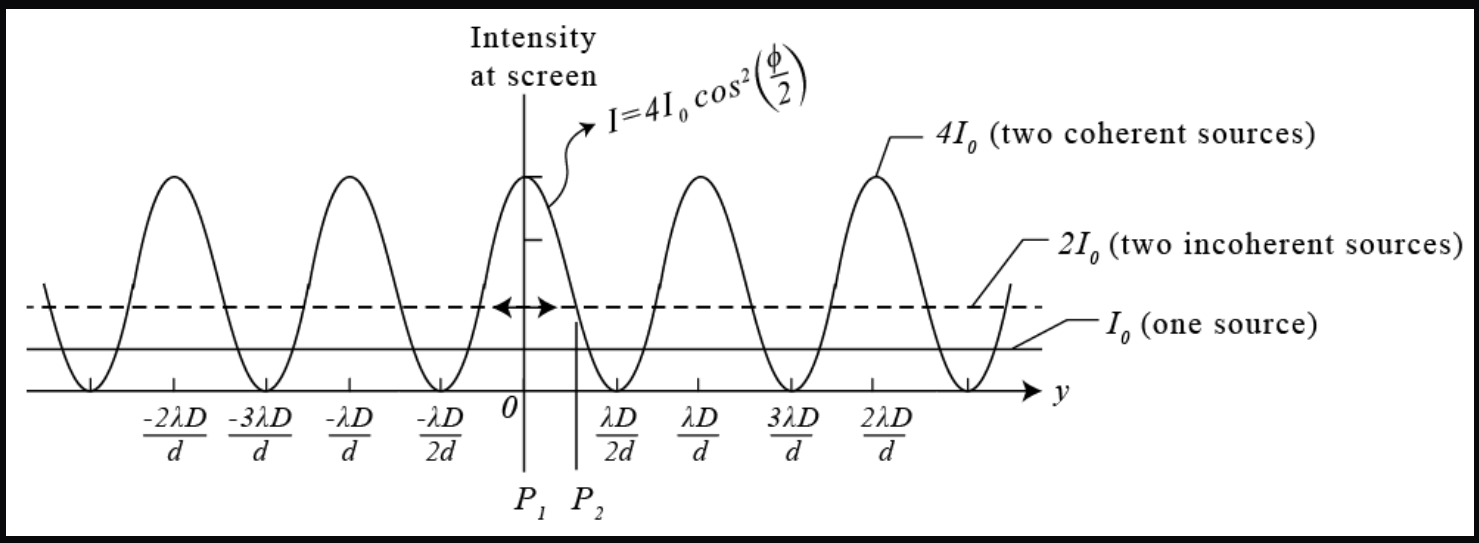
The area of the region bounded by the curves $y=9x^2$ and $y=5x^2+4$ (in sq units) is K/3, find K.

## Answers & Solutions

**Q.1 Answer:**

—

**Solution:**




$$\left.\begin{aligned} I_1 &= 4I_0 \\ I_2 &= 2I_0 \end{aligned}\right\}$$

$$y=\frac{1}{4}\left(\frac{\lambda D}{d}\right)$$

$$\Delta x=\frac{yd}{D}=\frac{\lambda}{4}$$

$$\Delta\phi=\frac{2\pi}{\lambda}\cdot\frac{\lambda}{4}=\frac{\pi}{2}$$

$$I=4I_0\cos^2\left(\frac{\pi}{4}\right)=2I_0$$

**Q.2 Answer:**

$\sqrt{2}-1$

**Solution:**

**If a body falls freely from rest at $A$. Then, time taken to travel from $A$ to $B$ (time taken)**

$t_1=\sqrt{\frac{2h}{8}}$

**Time taken to travel from $B$ to $C$ (time taken)**

$$t_2 = \sqrt{\frac{4h}{8}} - \sqrt{\frac{2h}{8}}$$

$$\frac{t_2}{t_1} = \sqrt{2} - 1$$

**Q.3 Answer:**

$-16.7$ cm and -0.67

**Solution:**

**Using mirror equation, we have,**

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$\Rightarrow \quad \frac{1}{v} = \frac{1}{(-10)} - \frac{1}{(-25)}$$

$$[\because \text{ Given } f = -10 \text{ cm}, u = -25 \text{ cm}]$$

$$\Rightarrow \quad v = -16.7 \text{ cm}$$

**or**

**and magnification,**

$$m = -\left(\frac{v}{u}\right) = \frac{-(-16.7)}{-25} = -0.67$$

**Q.4 Answer:**

increases by $2\%$

**Solution:**

**Using ideal gas equation,**

$$\mathrm{PV} = \mathrm{nRT}$$

**Volume is constant,**

$$\frac{P}{T} = \frac{nR}{V} = \text{ constant}$$

$$P \propto T$$

**If pressure increase by $2\%$ than temperature is also increase by $2\%$**

**Q.5 Answer:**

$5b - 4a$

**Solution:**

**Let the initial length of the string be " ?.**

**Tension $\propto$ Elongation of the string**

$$\frac{T_1}{T_2} = \frac{a-l}{b-l} \Rightarrow \frac{4}{5} = \frac{a-l}{b-l}$$

$$4b - 4l = 5a - 5l$$

$$l = 5a - 4b$$

Again, $\quad \dfrac{T_1}{T_3} = \dfrac{a-l}{x-l}$

$x =$ length of the string at tension $T(= 9\ \text{N})$

$$\frac{4}{9} = \frac{a-(5a-4b)}{x-(5a-4b)}$$

$$\frac{4}{9} = \frac{-4a+4b}{x-(5a-4b)} \Rightarrow \frac{4}{9} = \frac{4(-a+b)}{x-(5a-4b)}$$

$$\Rightarrow x-(5a-4b) = 9a+9b$$

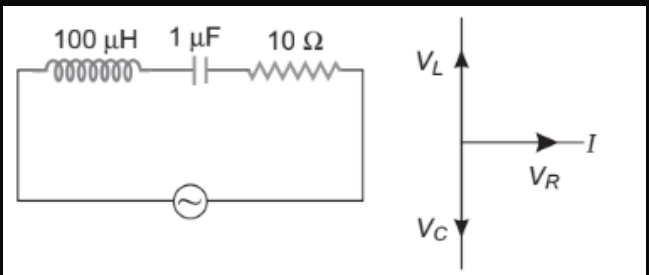
$$x = -9a+9b+5a-4b$$

$$= 5b-4a$$

**Q.6 Answer:**

series R-C circuit

**Solution:**




**Impedance,** $Z = \sqrt{(X_L \sim X_C)^2 + R^2}$

**or** $\quad Z = \sqrt{\left(\omega L \sim \frac{1}{\omega C}\right)^2 + R^2}$

**Inductive reactance**

$$X_L = \omega L = 70 \times 10^3 \times 100 \times 10^{-6}$$

$$= 7\Omega$$

**Capacitance reactance**

$$X_C = \frac{1}{\omega C} = \frac{1}{70 \times 10^3 \times 1 \times 10^{-6}}$$

$$= \frac{1}{7 \times 10^{-2}} = \frac{10^2}{7} = \frac{100}{7}$$

**As** $\quad X_C > X_L$

**Hence, circuit behave like as** $R - C$ **circuit.**

**Q.7 Answer:**

1,0

**Solution:**

$A(0), B(1)$

$(\mathrm{A} \cdot \overline{\mathrm{B}} + \overline{\mathrm{A}} \cdot \mathrm{B}) = (0.0 + 1.1) = 1$

$\mathrm{A} \cdot \overline{\mathrm{B}} + \overline{\mathrm{A}} \cdot \mathrm{B} = (0.1 + 1.0) = 0$

$(1, 0)$

**Q.8 Answer:**

$\frac{5}{2}$

**Solution:**

As, $P = \frac{a^{1/2} \cdot b^{1/2} \cdot d^{\alpha}}{c^{1/2}}$

So $\%$ error in $P = \frac{\Delta P}{P} \times 100$

$$= \frac{1}{2}\left(\frac{\Delta a}{a}\right) \times 100 + \frac{1}{2}\left(\frac{\Delta b}{b} \times 100\right) + \alpha\left(\frac{\Delta d}{d} \times 100\right) + \frac{1}{2}\left(\frac{\Delta c}{c} \times 100\right)$$

$\Rightarrow \%$ error in $P = \frac{1}{2} \times \%$ error in $a + \frac{1}{2} \times \%$ error in $b + \frac{1}{2} \times \%$ error in $c + \alpha \times \%$ error in $d$

Now given,

\% Error in $P = 2\%$

$\%$ Error in $a = \%$ error in $b$

$= \%$ Error in $c = \%$ error in $d = 0.5\%$

Substituting these values in Eq. (i), we get

$$2 = \frac{1}{2} \times 0.5 + \frac{1}{2} \times 0.5 + \frac{1}{2} \times 0.5 + \alpha \times 0.5$$

$$\Rightarrow 2 = \frac{1}{2}\left(\frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \alpha\right) \Rightarrow 4 = \frac{3}{2} + \alpha$$

or $\alpha = 4 - \frac{3}{2} = \frac{5}{2}$

$\therefore \alpha$ is $5/2$.

**Q.9 Answer:**

$v_1$

**Solution:**

Key Idea Root mean square velocity of any gas is given by

$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$

Where, $M$ = molecular weight.

As, rms speed of any gas is depends on the temperature of gas and molecular weight of gas.

In a mixture of gases (say $N_2$ and $O_2$ ) at a constant temperature, rms speed is independent to the quantity (moles) of gas present in the gas mixture. So, in this problem the rms speed of $O_2$ in vessel $C$ is $v_1$.

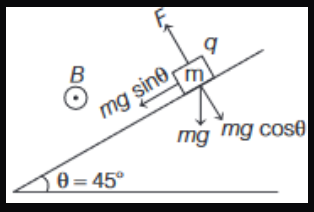
Hence, the correct option is (b).

**Q.10 Answer:**

5 s

**Solution:**

Given inclination of plane,




Mass of block, $m = 20\ \text{g} = 0.02\ \text{kg}$

charge on the block, $q = 4\text{mC} = 4 \times 10^{-3}\text{C}$

Magnetic field, $B = 1\ \text{T}$

Magnetic force on the charge particles, $F = Bqv$

Particle will leave the inclined plane, when

$F = mg\cos\theta \Rightarrow Bqv = mg\cos\theta \Rightarrow v = \frac{mg\cos\theta}{qB}$

Time taken to reached at the velocity $v$ is given by

$v = 0 + g\sin\theta t \quad [\because u = 0, a = g\sin\theta]$

$$t = \frac{v}{g\sin\theta} = \frac{mg\cos\theta}{qB \cdot g\sin\theta} = \frac{m\cot\theta}{qB}$$

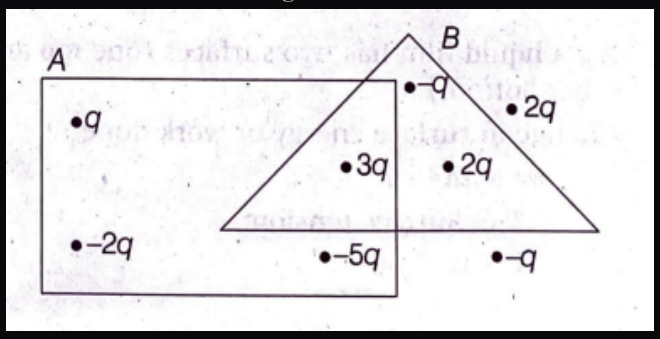
$$t = \frac{0.02\cot 45^\circ}{4 \times 10^{-3} \times 1} = 5\ \text{s} \quad [\because \cot 45^\circ = 1]$$

**Q.11 Answer:**

$-\frac{3}{4}$

**Solution:**

The distribution of charges on two Gaussian surfaces $A$ and $B$ are shown below




Flux through a closed surface, $\phi = \frac{q_{\text{net}}}{\varepsilon_0}$

$q_{\text{net}}$ = net charge enclosed by the surface

Here, $(q_{\text{net}})_A$ = charge enclosed by surface

$A = q + 3q - 2q - 5q = -3q$

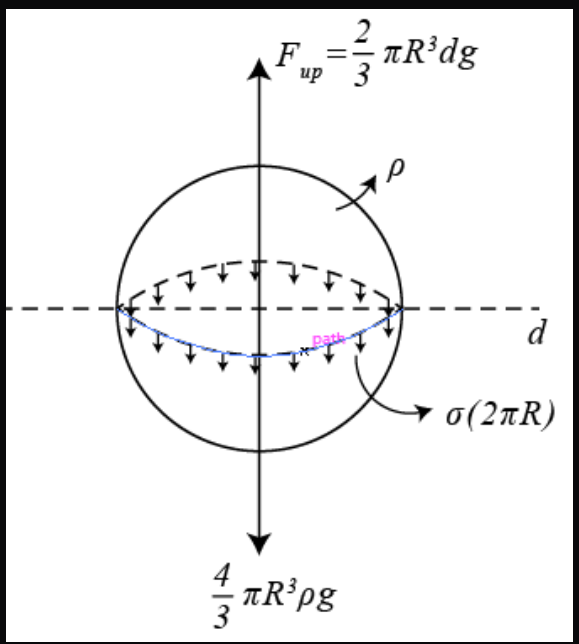
And $(q_{\text{net}})_B = -q + 3q + 2q = 4q$

So, $\frac{\phi_A}{\phi_B} = \frac{-3q/\varepsilon_0}{4q/\varepsilon_0} = \frac{-3}{4}$

**Q.12 Answer:**

$\sqrt{\frac{3\sigma}{g(d-2\rho)}}$

**Solution:**



$$\frac{2}{3}\pi R^3 dg = \frac{4}{3}\pi R^3 \rho g + \sigma(2\pi R)$$
$$\frac{R^2}{3}\{dg - 2\rho g\} = \sigma$$
$$R^2 = \frac{3\sigma}{g(d-2\rho)}$$
$$R = \sqrt{\frac{3\sigma}{g(d-2\rho)}}$$

**Q.13 Answer:**

4.5

**Solution:**

$m = 4\ \text{kg}, p = 8\ \text{kg m/s}$

$F = 0.2\ \text{N}, t = 10\text{sec}$

**From Newton's IInd law**

**Force = rate of change of momentum**

$$F = \frac{p_2 - p_1}{t}$$
$$0.2 = \frac{mv - 8}{10}$$
$$2 = 4 \times v - 8$$
$$4v = 10$$
$$\Rightarrow \quad v = \frac{5}{2}\ \text{m/s}$$
$$\therefore \quad \text{Final momentum } p_2 = mv = 4 \times \frac{5}{2}$$
$$= 10\ \text{kg m/s}$$

**Q.14 Answer:**

III and IV only

**Solution:**

**The maximum kinetic energy of the photoelectrons depends on the frequency of incident light and does not depend on the intensity of light.**

$K_{max} = h\nu - \phi$

**Therefore, statement III and IV are correct and I is incorrect.**

**Photoelectric effect happens when insident radiation ha frequency greater than minimum frequency called as threshold frequency. Therefore, statement II is incorrect.**

**Q.15 Answer:**

$8\ \text{ms}^{-2}$

**Solution:**

**Block's weight (downward),** $w = 2 \times 10 = 20\ \text{N}$

**Vertical component of applied force (upwards),**

$F_V = F\sin\theta = 90 \times 3/5 = 54\ \text{N}$

**Maximum frictional force,**

$F_r = \mu F\cos\theta = 0 \cdot 25 \times 90 \times 4/5$

$= 18\ \text{N}$

$\therefore$ Net vertical force, $F_{\text{net}} = F_V - (F_r + w) = ma$

$\therefore \quad F_{\text{net}} = 54 - 18 - 20 = 16\ \text{N} = ma$

$\Rightarrow \quad a = \frac{16}{2} = 8\ \text{m/s}^2$

**Q.16 Answer:**

$-27.2\text{eV}$

**Solution:**

Mass of the electron, $\text{m} = 2\ \text{m}$

Energy of electron, $E \propto \frac{\text{m}}{\text{n}^2}$ For first orbit, $\text{n} = 1$

$$E = \frac{2 \times (-13.6)}{1}$$
$$= -27.2\text{ev}$$

**Q.17 Answer:**

$1 \times 10^{-6}$

**Solution:**

$l = 50\ \text{cm} = 50 \times 10^{-2}\ \text{m}$

$A = 1\ \text{mm}^2 = 1 \times 10^{-4}\ \text{m}^2$

$i = 4\ \text{A}, V = 2\text{volt}$

$\therefore$ Resistance of wire, $R = \frac{V}{i} = \frac{2}{4} = 0.5\Omega$

$$\therefore \text{ Resistiviry, } \rho = R\frac{A}{l} = 0.5 \times \frac{1 \times 10^{-6}}{50 \times 10^{-2}}$$
$$= \frac{0.5 \times 1 \times 10^{-4}}{50}$$
$$= 1 \times 10^{-4}\Omega - \text{fm}$$

**Q.18 Answer:**

Visible red light

**Solution:**

Ultraviolet light is electromagnetic radiation with a wavelength shorter than that of visible light, but longer than X-rays.

$\therefore \quad \lambda_{\text{red}} > \lambda_{\text{blue}} > \lambda_{\text{UV}} > \lambda_{\text{X-ray}}$

Hence, visible red light has longest wavelength.

**Q.19 Answer:**

80 N

**Solution:**

Given, radius of Earth and Mars are $6400\ \text{km}$ and $3200\ \text{km}$, respectively.

Mass of Earth $= 10$ (Mass of Mars)

$M_e = 10M_m$

We know that, acceleration due to gravity at surface of a planet is

$g = \frac{GM}{R^2}$,

where, $M =$ mass of planet

and $R =$ radius of planet.

Taking the ratio of acceleration due to gravity of Mars to Earth,

$$\frac{g_m}{g_e} = \frac{\frac{GM_m}{R_m^2}}{\frac{GM_e}{R_e^2}} = \frac{M_m}{M_e} \times \left(\frac{R_e}{R_m}\right)^2$$

By substituting the values, we get

$$\frac{g_m}{g_e} = \frac{1}{10} \times \left(\frac{6400}{3200}\right)^2 = \frac{1}{10} \times 4$$

$$g_m = \frac{2}{5} g_e$$

**Given, the weight on earth's surface,**

$w_e = mg_e = 200 \text{ N}$

**$\therefore$ Weight on the Mars surface,**

$w_m = mg_m = \frac{2}{5}(mg_e) = \frac{2}{5} \times 200 \text{ N}$

$= 80 \text{ N}$

**Q.20 Answer:**

$1.5 \text{ m}, -3 \text{ ms}^{-1}$

**Solution:**

**When block strikes the spring, the kinetic energy of block converts into potential energy of spring ie,**

$\frac{1}{2}mv^2 = \frac{1}{2}kx^2$

**where $x$ is compression in spring.**

$x = \sqrt{\frac{mv^2}{k}}$

$= \sqrt{\frac{25 \times (3)^2}{100}} = \sqrt{\frac{9}{4}} = \frac{3}{2} = 1.5 \text{ m}$

**When block returns to the original position, again potential energy converts into kinetic energy of the block, so velocity of the block is same as before but its sign changes as it goes to mean position.**

**Hence,** $v = -3 \text{ ms}^{-1}$.

**Q.21 Answer:**

—

**Solution:**

**According to question,**

$B_1 = \frac{\mu_0 I R^2}{2(R^2+r^2)^{3/2}}$

**Here,** $r = R\sqrt{3}$

$$\therefore \quad B_1 = \frac{\mu_0 I R^2}{2\left[R^2 + (R\sqrt{3})^2\right]^{3/2}} = \frac{\mu_0 I R^2}{2\left[4R^2\right]^{3/2}} = \frac{\mu_0 I R^2}{16R^3}$$

$$B_1 = \frac{\mu_0 I}{16R} \quad \ldots(i)$$

**and** $B_2 = \frac{\mu_0 I}{2R}$ **...(ii)**

**[magnetic field at centre]**

**Now, dividing Eq. (i) by Eq. (ii), we get**

$\therefore \quad \frac{B_1}{B_2} = \frac{\mu_0 I/16R}{\mu_0 I/2R} = \frac{1}{8}$

**Q.22 Answer:**

—

**Solution:**

**Given,**

**Young's modulus of material** $= 2 \times 10^{11} \text{ N/m}^2$

**Elastic limit or stress** $= 1 \times 10^{\theta} \text{N/m}^2$

**Length of the wire** $= 1 \text{ m}$

**We know that,**

$Y = \frac{\text{Stress}}{\text{Strain}} \Rightarrow Y = \frac{\text{Stress}}{\frac{\Delta L}{l}}$

**Here,** $l$ **= original length of the wire**

$\Delta L =$ charge in the length of the wire

$$\Delta L = \frac{\text{Stress } \times l}{Y} = \frac{1 \times 10^{\theta} \times 1}{2 \times 10^{11}}$$
$$= 0.5 \times 10^{-3} = 0.5 \text{ mm}$$

**Q.23 Answer:**

—

**Solution:**

Given, velocity, $v = 6t - 3t^2$

As we know that,

$v = \frac{dx}{dt}$

Here, $x$ is the displacement of the particle.

Now, $dx = vdt$

Integrate on the both sides, limit $t = 0$ to $t = 2$, we get

$$\therefore \quad x = \int_0^2 vdt = \int_0^2 \left(6t - 3t^2\right) dt$$
$$= \left[\frac{6t^2}{2}\right]_0^2 - \left[\frac{3t^3}{3}\right]_0^2 = \left[3t^2\right]_0^2 - \left[t^3\right]_0^2$$
$$= \left[3(2)^2 - 3(0)^2\right] - \left[(2)^3 - (0)^2\right]$$
$$= [12 - 0] - [8 - 0] = 12 - 8 = 4 \text{ m}$$

$$\text{Average velocity, } v_{\text{avg}} = \frac{\text{Total displacement}}{\text{Total time taken}}$$
$$= \frac{4}{2} = 2 \text{ m/s}$$

Hence, the average velocity of the object between $t = 0$ to $t = 2$ s is $2 \text{ m/s}$.

**Q.24 Answer:**

—

**Solution:**

$N = 1000, A = 500 \text{ cm}^2 = 500 \times 10^{-4}$
$= 5 \times 10^{-2} \text{ m}^2$
$B = 2 \times 10^{-5} \text{ Wb/m}^2, \theta_1 = 0^\circ,$
$\theta_2 = 180^\circ, \Delta t = 0.2 \text{ s}$

Initial flux linked with coil,

$\phi_1 = NBA \cos \theta_1$
$= NBA \cos 0^\circ$
$= NBA$

Final flux, $\phi_2 = NBA \cos 180^\circ$
$= NBA(-1) = -NBA$

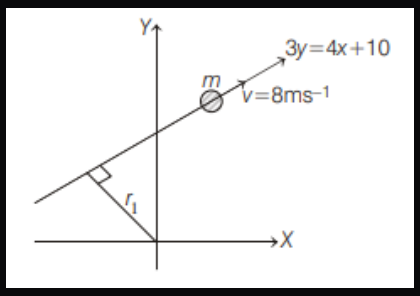
Change in flux, $\phi = \phi_2 - \phi_1$
$= -NBA - (NBA)$
$= -2NBA$

$\therefore$ Induced emf, $e = \frac{-\Delta\phi}{\Delta t} = -\frac{(-2NBA)}{\Delta t}$
$= \frac{2NBA}{\Delta t}$
$= \frac{2 \times 1000 \times 2 \times 10^{-5} \times 5 \times 10^{-2}}{0.2}$
$= 10 \times 10^{-3} \text{ V}$
$= 10\text{mV}$

**Q.25 Answer:**

—

**Solution:**




**Angular momentum** $= mvr_1$

$= 15 \times 8 \times \left(\frac{10}{\sqrt{3^2+4^2}}\right) = 15 \times 8 \times 2 = 240 \text{ kg} - \text{m}^2 \text{ s}^{-1}$

**Q.26 Answer:**

Triamminediaqua chlorido chromium (III) ch!oride.

**Solution:**

**The IUPAC name of the compound** $[\text{Cr}(\text{NH}_3)(\text{H}_2\text{O})_2\text{Cl}_{\text{Cl}}\text{Cl}_2$ **is triamminediaqua chlorido chromium (III) chloride.**

**Q.27 Answer:**

(iii) > (iv) > (ii) > (i)

**Solution:**

**(i)** $\text{CH}_3 - \text{CH}_2 - \text{COOH}$

**(ii)** $\text{CH}_3 - \text{COOH}$

**(iii)** $\text{C}_6\text{H} - \text{COOH}$

**(iv)** $\text{C}_6\text{H}_4 - \text{CH}_2 - \text{COOH}$

**Die to** $-I$ **effect (iii) is most acidic and order will be (iii)** $>$ **(iv)** $>$ **(ii)** $>$ **(i).**

**Q.28 Answer:**

Al $<$ Ga $<$ In $<$ TI

**Solution:**

**The stability of +1 oxidation state among Al,**

$\text{Ga}$**, In and** $\text{Tl}$ **increases in the sequence**

$\text{Al} < \text{Ga} < \text{In} < \text{Tl}$

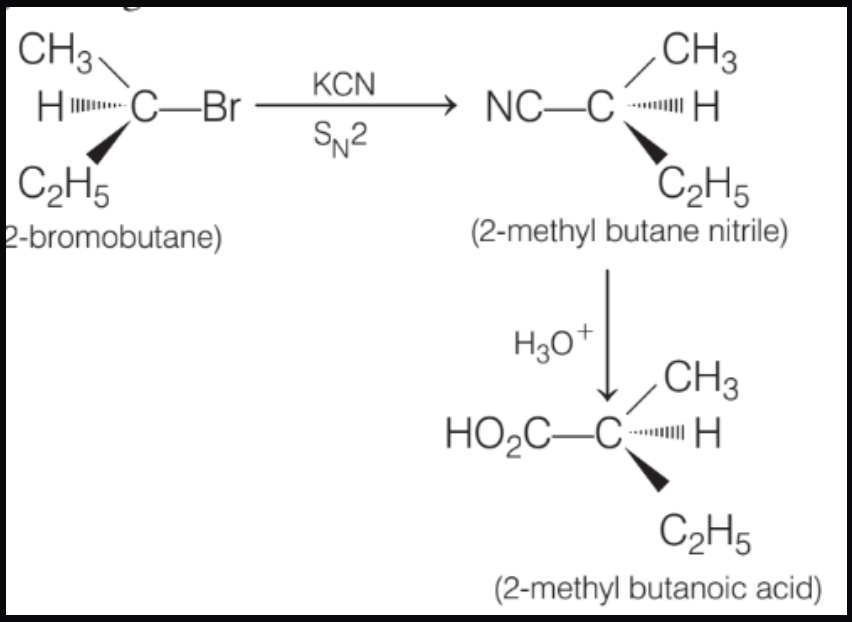
**This is due to inert pair effect due to which on moving down the group, the stability of +3 oxidation state decreases while stability of +1 oxidation state increases.**

**Q.29 Answer:**



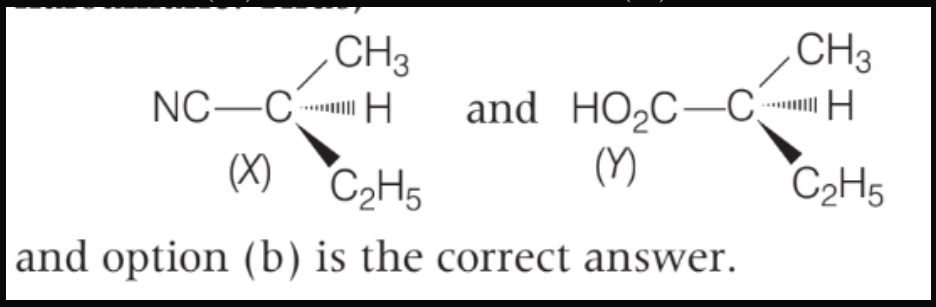

**Solution:**

**The given reaction occurs as follows :**




**In the above reaction, $2^{\circ}$ haloalkane reacts with $\mathrm{KCN}$ via $\mathrm{S_N2}$ mechanism due to less steric hinderance and we get 2-methyl butane nitrile $(X)$ with inversion in structure.**

**On hydrolysis of $(X)$, we get 2-methyl butanoic acid $(Y)$ having inverted structure as that of haloalkane. Thus,**




and option (b) is the correct answer.

**Q.30 Answer:**

$\mathrm{C_2H_5OH}$

**Solution:**

$$\underset{(X)}{\mathrm{C_2H_5OH}} + [\mathrm{O}] \xrightarrow[\text{in } \mathrm{CH_2Cl_2}]{\mathrm{PCC}} \underset{(Y)}{\mathrm{CH_3CHO}}$$

$$\underset{(Y)}{\mathrm{CH_3CHO}} + 4\mathrm{NaOH} + 3\mathrm{I_2} \longrightarrow \mathrm{CHI_3} + \mathrm{HCOONa} + 3\mathrm{H_2O} + 3\mathrm{NaI}$$

**Q.31 Answer:**

II, III only

**Solution:**

**(i) The ionic radius of lanthanoids follow the following relation:**

**Ionic radius** $\propto \frac{1}{\text{atomic number}}$

**Atomic number** $(Z)$ **for** $\mathrm{Pr} = 59$

**Atomic number** $(Z)$ **for** $\mathrm{Sm} = 62$

**Atomic number** $(Z)$ **for** $Dy = 66$

**and has ionic radii for positive ions as given below:**

$\mathrm{Pr}^{3+} = 1.013\mathring{A}$

$\mathrm{Sm}^{3+} = 0.964\mathring{A}$

$\mathrm{Dy}^{3+} = 0.908\mathring{A}$

Thus, statement (i) is not correct.

(ii) Element Eu $(Z = 63)$ can exist in $(+)2$ and $(+)3$ oxidation state.

$\therefore (+)$ 3oxidation state is the most common oxidation state for lanthanoids.

Thus, (+) 2oxidation state oxides oxidises to (+) 3 easily and $\mathrm{Eu}^{2+}$ can act as strong reducing agent.

Hence, the statement (II) is correct.

(iii) The electronic configuation for

$(\mathrm{Pu}), (Z = 94) = 5f^6 6d^0 7s^2$

Therefore, it can use maximum of $6 + 2 = 8$ electrons.

Thus, can show +7 oxidation state.

Thus, statement (III) is also correct.

Hence, option (d) is the correct answer.

**Q.32 Answer:**

$\mathrm{NH_4NO_3}$

**Solution:**

$\mathrm{NH_4NO_3}$ will evolve $\mathrm{NH_3}$ pungent smelling gas with $\mathrm{NaOH}$.

- $\mathrm{NO_2}$ brown gas with $\mathrm{con.H_2SO_4}$

- No reaction with water

Thus, $\mathrm{NH_4NO_3}$ can be used to label all three beakers.

(c) $\mathrm{(NH_4)_2\,CO_3}$ will evolve pungent smelling gas $\mathrm{NH_3}$ with $\mathrm{NaOH}$

- Effervescence of $\mathrm{CO_2}$ gas with con. $\mathrm{H_2SO_4}$

- No reaction with water.

Thus, $\mathrm{(NH_4)_2\,CO_3}$ can be used to label all three beakers.

Thus, options $(a, c)$ are correct.

**Q.33 Answer:**

$\mathrm{Cr^{2+}}$ is a reducing agent

**Solution:**

$\mathrm{Cr^{2+}} \Rightarrow [\mathrm{Ar}]3d^4, E^{\circ}_{\mathrm{Cr^{3+}/Cr^{2+}}} = -0.41\ \mathrm{V}$

$\mathrm{Mn^{3+}} \Rightarrow [\mathrm{Ar}]3d^3, E^{0}_{\mathrm{Mn^{3+}/Mn^{2+}}} = +1.51\ \mathrm{V}$

So, reducing power $\mathrm{Cr^{2+}} > \mathrm{Mn^{2+}}$

Why is $\mathrm{Cr^{2+}}$ reducing and $\mathrm{Mn^{3+}}$ oxidising when both have $\mathrm{d^4}$ configuration.

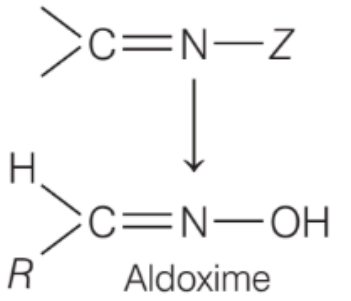
$\mathrm{Cr^{2+}}$ is reducing as its configuration changes from $d^4$ to $d^3$, the latter having a half-filled $t_{2g}$ level (see Unit 9). On the other hand, the change from $\mathrm{Mn^{2+}}$ to $\mathrm{Mn^{3+}}$ results in the half-filled $\left(d^5\right)$ configuration which has extra stability.

**Q.34 Answer:**

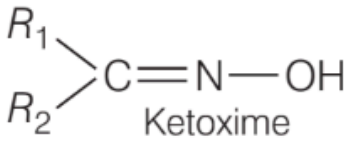
$\mathrm{OH \quad NHCONH_2}$

**Solution:**

(c) Oxime are



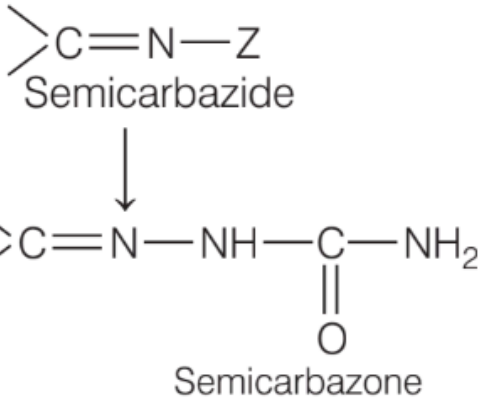
Aldoxime



Ketoxime

Here, $Z = OH$

Semicarbazone formed by



Semicarbazide

Semicarbazone

Here, $Z = -NH-\underset{\underset{O}{\|}}{C}-NH_2$

Thus, option (c) is correct.

**Q.35 Answer:**

red

**Solution:**

**When a compound absorbs a certain wavelength (i.e., $490\ \text{nm} - 500\ \text{nm}$ ) from the visible light which corresponds to the blue-green light in the visible spectrum, the colour or light transmitted by it does not contain the colour of absorbed radiation and thus shows the complementary colour, i.e., red.**

**Q.36 Answer:**

iii, iv

**Solution:**

**(i) Electromeric effect is a temporary effect, it takes place only in the presence of reagent.**

**(ii) Hyper-conjugation is a permanent effect which allows delocalised sigma electrons to adjacent empty $\pi$ or $p$-orbital.**

**Q.37 Answer:**

$2 < 1 < 3 < 4$

**Solution:**

The correct order is $2 < 1 < 3 < 4$. Since, more the number of $2^{\circ}$ - NH- group, more is the basic nature.

(1) Thus, 3 and 4 are more basic than 1 and 2 .

(ii) Between 3 and 4,4 has one two - $NH_2$ group while 3 has only one $-NH_2$ group. $\therefore$ 4 is more basic than 3

(iii) Between 1 and 2,1 has one $-NH_2$ group with no resonance, thus is more basic than 2.

**Q.38 Answer:**

$Li^{+2}$

**Solution:**

Hints: $\frac{1}{\lambda} = z^2 \cdot R_H \left[1/n_1^2 - 1/n_2^2\right]$

$\Rightarrow \frac{1}{\lambda} = (z)^2 \cdot R_H \left\{\frac{1}{1} - \frac{1}{4}\right\} = \frac{3}{4} R_H z^2$

$\therefore \lambda \propto 1/z^2$

Hence, for shortest $\lambda$, z must be maximum, which is for $Li^{+2}$.

**Q.39 Answer:**

$B, C, D$

**Solution:**

The species which contain free $-CHO$ group are reducing sugars.

(A) Sucrose is a non-reducing sugar.

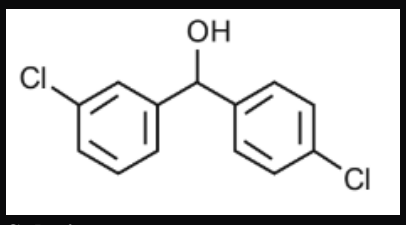
(B) Maltose is a reducing sugar.

(C) Lactose is a reducing sugar.

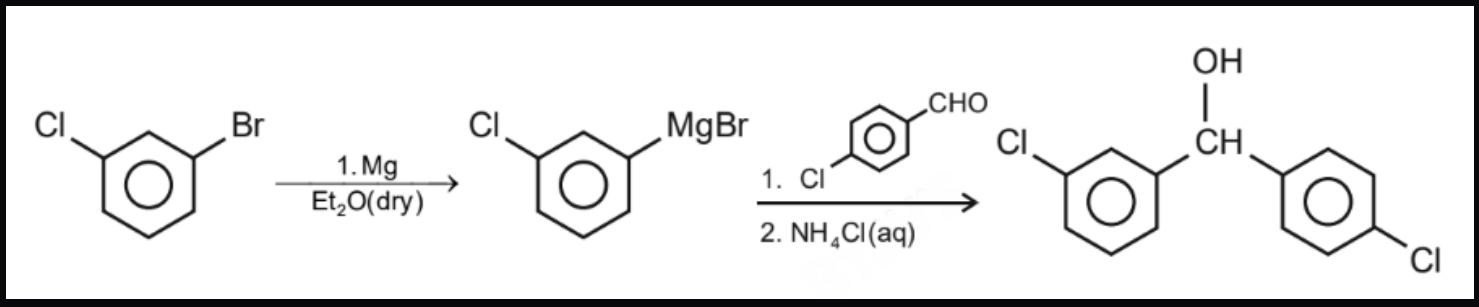
(D) Fructose is a reducing sugar.

Hence, option (4) is the correct answer.

**Q.40 Answer:**



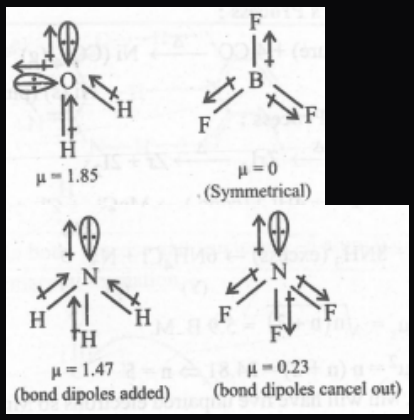

**Solution:**




**Q.41 Answer:**

A - IV, B - I, C - III, D - II

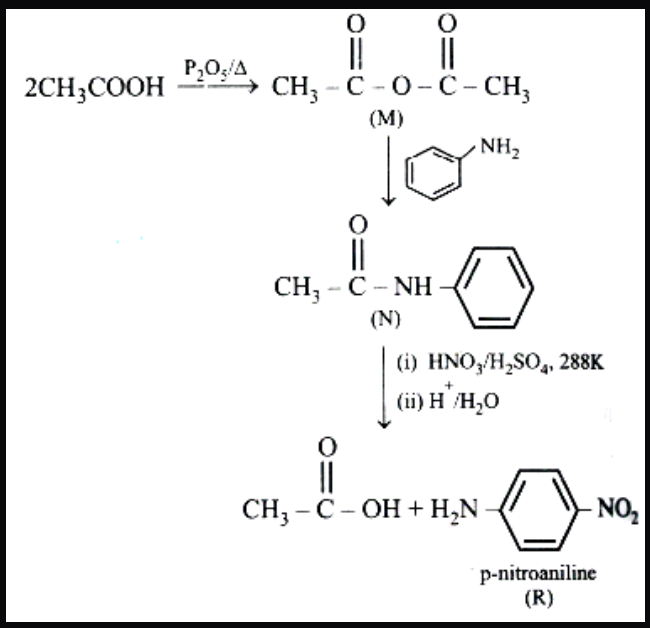
**Solution:**




**Q.42 Answer:**

p-Nitroaniline

**Solution:**




**Q.43 Answer:**

increases from left to right and decreases from top to bottom

**Solution:**

**More the electropositivity of central atom, the more basic is the oxide and more electronegative central atom, more acidic is the oxide.**

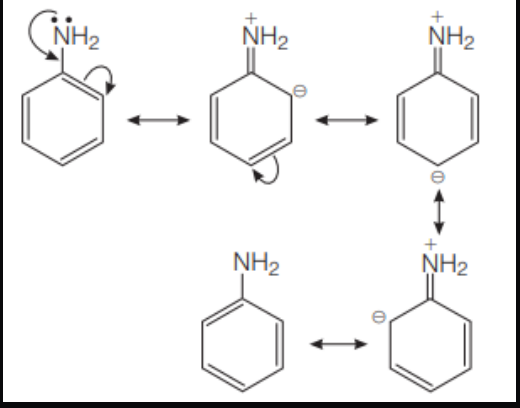
**As we know, the electronegativity is the property of the element to attract electrons pairs. Hence, acidic character of the oxides "decreases down the group" because the electronegativity decreases down the group. Electronegativity decreases down the group because the "distance between the nucleus and valences shell electron increases. Hence, acidic nature of oxides increases left to right and decreases from top to bottom.**

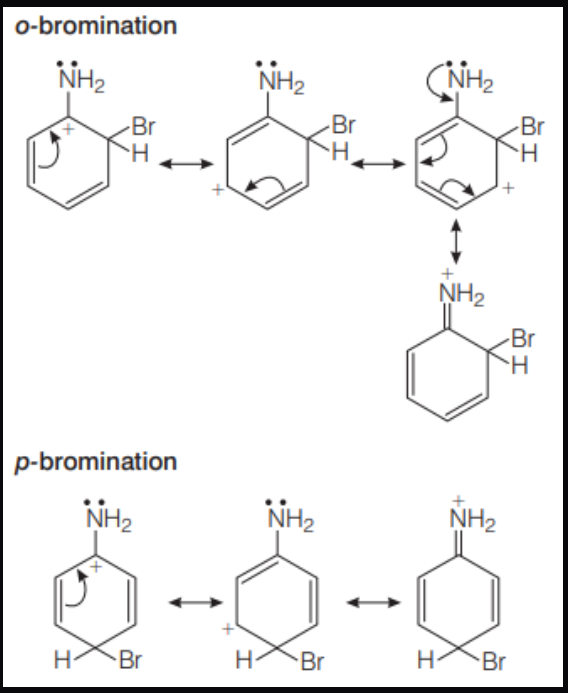
**Q.44 Answer:**

Both $(A)$ and $(R)$ are correct, $(R)$ is the correct explanation of $(A)$

**Solution:**

**The $-NH_2$ group of aniline is a very strong electron donor (+ $M$ effect), hence it activates the benzene ring thoroughly and the electrophilic aromatic substitutions on the benzene ring are very easy to take place at ortho and para-positions.**






**Q.45 Answer:**

$A > B > C$

**Solution:**

Ionisation potential of $A$, $B$ and $C$ elements decreases with the ease of removal of electron from $A$ to $C$. Atomic size increase from $A$ to $C$, hence ionisation potential decrease in the same order.

So, order of ionisation potential is $A > B > C$

**Q.46 Answer:**

—

**Solution:**

$\because$ Number of milli equivalents of NaOH

$(n') = NV = 0.1 \times 20 = 2$

$\therefore$ Number of equivalent $= 0.002$

For Neutralisation Number of equivalent of base (NaOH) Number of equivalent of acid

$0.002 = \frac{W}{E} = \frac{0.126}{E}$

$\therefore \quad E = \frac{0.126}{0.002} = 63$

**Q.47 Answer:**

—

**Solution:**

Salts such as $\mathrm{NaCl}, \mathrm{HgCl_2}, \mathrm{CuCl_2}$ are soluble in water while salts such as $\mathrm{Hg_2Cl_2}, \mathrm{CuCl}$ and $\mathrm{AgCl}$ are insoluble in water.

**Q.48 Answer:**

—

**Solution:**

(d) $A + B \longrightarrow P$

According to rate law expression

$$\text{Rate } = k[A]^{\alpha}[B]^{\beta} (K \text{ is constant }) \ldots (i)$$

Taking ratio of (1) and (2) entry,

$$\frac{4 \times 10^{-2}}{2 \times 10^{-2}} = \frac{[0.04]^{\beta}[0.02]^{\alpha}}{[0.02]^{\beta}[0.02]^{\alpha}}$$

$$2 = [2]^{\beta}$$

$$[2]^1 = [2]^{\beta}$$

$$\beta = 1$$

Taking ratio of (2) and (3) entry,

$$\frac{8 \times 10^{-2}}{4 \times 10^{-2}} = \frac{[0.04]^{\alpha}[0.04]^{\beta}}{[0.02]^{\alpha}[0.04]^{\beta}}$$

$$[2] = [2]^{\alpha} \Rightarrow \alpha = 1$$

Put value of $\alpha$ and $\beta$ in (1) entry

$$2 \times 10^{-2} = k[A]^1[B]^1$$

$$k = \frac{2 \times 10^{-2}}{0.02 \times 0.02}$$

$$= \frac{2 \times 10^{-2} \times 10^4}{4} = \frac{100}{2} = 50$$

**Q.49 Answer:**

—

**Solution:**

The atomisation enthalpy of $\mathrm{CH_4}$ is $1660 \text{ kJ mol}^{-1}$.

$\mathrm{CH_4}(g) \longrightarrow \mathrm{C}(g) + 4\mathrm{H}(g); \quad \Delta_a H = 1660 \mathrm{kJ mol^{-1}}$

$\mathrm{CH_4} \xrightarrow{15} \mathrm{CH_3} \xrightarrow{30} \mathrm{CH_2} \xrightarrow{45} \mathrm{CH}$

Mean $\mathrm{C-H}$ bond enthalpy $= \frac{1660}{4}$

$\therefore \quad 415 = 15 + 30 + 45 + x$

$\Rightarrow \quad x = 415 - 90 = 325 \text{ kJ mol}^{-1}$

**Q.50 Answer:**

—

**Solution:**

Bond order : It shows the number of chemical bonds present between a pair of atoms.

| Element | $\mathrm{N_2^-}$ | $\mathrm{NO^-}$ | $\mathrm{C_2^-}$ | $\mathrm{N_2^+}$ | $\mathrm{C_2^{2-}}$ | $\mathrm{CN^+}$ |
|---|---|---|---|---|---|---|
| Bond order | 2.5 | 2 | 2.5 | 2.5 | 3 | 2 |

Here, 3 ions are having a bond order of 2.5.

**Q.51 Answer:**

3

**Solution:**

Consider the numbers $a/2, a/2, b/3, b/3, b/3, c/4, c/4, c/4, c/4$ using A.M. $\geq$ G.M. we get

$\frac{a+b+c}{9} \geq \left(\frac{a^2b^3c^4}{2^{10}3^3}\right)^{1/9}$

$\Rightarrow$ maximum value of $a^2b^3c^4$ is $2^{10} \times 3^3$

Hence $x = 10$ and $y = 3$

$\therefore \log_{10}(x^y) = \log_{10}(10^3) = 3$

**Q.52 Answer:**

5040

**Solution:**

We have $F(x) = \frac{x^3}{3} + (a-3)x^2 + x - 13$

$\therefore$ For $\mathrm{F(x)}$ to have negative point of local minimum, the equation $\mathrm{F'(x) = 0}$ must have two distinct negative roots.

Now, $F'(x) = x^2 + 2(a-3)x + 1$

$\therefore$ Following condition(s) must be satisfied simultaneously.

(i) Discriminant $> 0$;

(ii) Sum of roots $< 0$;

(iii) Product of roots $> 0$

Now, $\mathrm{D} > 0$

$\Rightarrow 4(a-3)^2 > 4$

$\Rightarrow (a-3)^2 - 1 > 0$

$\Rightarrow (a-2)(a-4) > 0$

$\therefore a \in (-\infty, 2) \cup (4, \infty) \ldots (i)$

Also $-2(a-3) < 0$

$\Rightarrow \mathrm{a} - 3 > 0$

$\Rightarrow \mathrm{a} > 3 \ldots (ii)$

And product of $\mathrm{root(s)} = 1 > 0 \forall \mathrm{a} \in \mathrm{R}$

$\therefore$ (i) $\cap$ (ii) $\cap$ (iii)

$\Rightarrow a \in (4, \infty) \ldots (iii)$

Hence sum of value(s) of $a = 5 + 6 + 7 + \ldots\ldots\ldots + 100 = 5040$

**Q.53 Answer:**

$x - y + 7 = 0$

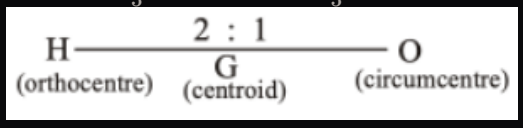
**Solution:**

Clearly circumcentre is $(0, 0)$

Now centroid G is $\left(\frac{5-13\cos\theta+13\sin\theta}{3}, \frac{12+13\sin\theta-13\cos\theta}{3}\right)$

G divides $\mathrm{HO}$ internally 2 : 1

$\therefore \frac{5-13\cos\theta+13\sin\theta}{3} = \frac{2(0)+1(\mathrm{h})}{3}$ and $\frac{12+13\sin\theta-13\cos\theta}{3} = \frac{2(0)+1(\mathrm{k})}{3}$




$\Rightarrow \sin\theta - \cos\theta = \frac{\mathrm{h}-5}{13}$ ...(1)

and $\sin\theta - \cos\theta = \frac{\mathrm{k}-12}{13}$ ...(2)

$\therefore$ From (1) and (2), we get

$\frac{\mathrm{h}-5}{13} = \frac{\mathrm{k}-12}{13}$

$\Rightarrow h - k + 7 = 0$

$\therefore$ **Locus of orthocentre** $\mathrm{H}(h, \mathrm{k})$ **is**

$x - y + 7 = 0$

**Q.54 Answer:**

Odd and is strictly decreasing in $(-\infty, \infty)$

**Solution:**

**We have** $g(t) = \frac{\pi}{2} - 2\cot^{-1}\left(3^{-t}\right)$

$\therefore g(-t) = \frac{\pi}{2} - 2\cot^{-1}\left(3^{t}\right) = \frac{\pi}{2} - 2\tan^{-1}\left(3^{-t}\right)\left(\text{As } \cot^{-1} x = \tan^{-1}\frac{1}{x}, x > 0\right)$

$= \frac{\pi}{2} - 2\left(\frac{\pi}{2} - \cot^{-1}\left(3^{-t}\right)\right) = -\frac{\pi}{2} + 2\cot^{-1}\left(3^{-t}\right)\left(\text{As } \cot^{-1} x + \tan^{-1} x = \frac{\pi}{2}, \forall x \in R\right)$

$= -g(t)$

**Hence** $g(-t) = -g(t)$

$\Rightarrow$ **g is an odd function.**

$\text{Also } \mathrm{g}'(\mathrm{t}) = \frac{-2.3^{-\mathrm{t}} \cdot \ln 3}{1 + \left(3^{-\mathrm{t}}\right)^2}$

$\therefore \mathrm{g}'(\mathrm{t}) < 0, \forall \mathrm{t} \in \mathrm{R}$

$\Rightarrow$ **g is strictly decreasing in** $(-\infty, \infty)$

**Q.55 Answer:**

5

**Solution:**

$\int \frac{x^2}{x^4 + 3x^2 + 9} dx = \frac{1}{2}\int \frac{x^2 + 3 + x^2 - 3}{x^4 + 9 + 3x^2} dx$

$= \frac{1}{2}\int \frac{d\left(x - \frac{3}{x}\right)}{\left(x - \frac{3}{x}\right)^2 + 9} + \frac{1}{2}\int \frac{d\left(x + \frac{3}{x}\right)}{\left(x + \frac{3}{x}\right)^2 - 3} = \frac{1}{6}\tan^{-1}\frac{x^2 - 3}{3x} + \frac{1}{4\sqrt{3}}\ln\left|\frac{x^2 - \sqrt{3}x + 3}{x^2 + \sqrt{3}x + 3}\right| + c$

$\therefore A = \frac{1}{6}, B = \frac{1}{4}$

$\therefore 12(A + B) = 5$

**Q.56 Answer:**

29

**Solution:**

We have, $n = 8, \bar{x} = 25$ and $\sigma = 5$

$\bar{x} = \dfrac{\Sigma x_i}{n}$

$\Rightarrow \quad \Sigma x_i = n\bar{x} = 8 \times 25 = 200$

$\Rightarrow$ In corrected $\Sigma x_i = 200$

and $\quad \sigma = 5$

$\Rightarrow \quad \sigma^2 = 25$

$\Rightarrow \dfrac{1}{n}\Sigma x_i^2 - (\text{ mean })^2 = 25$

$\Rightarrow \quad \dfrac{\Sigma x_i^2}{8} - 625 = 25$

$\Rightarrow \quad \Sigma x_i^2 = 5200$

Corrected $\Sigma x_i^2 = 5200 + 225 + 625 = 6050$

and corrected mean $= 200 + 15 + 25 = 240$

$\therefore$ Corrected variance $= \dfrac{6050}{10} - \left(\dfrac{240}{10}\right)^2$

$= 605 - (24)^2 = 605 - 576 = 29$

**Q.57 Answer:**

2

**Solution:**

$\vec{a} = 3\hat{i} - \hat{j} + \lambda\hat{k}$ and $\vec{b} = \lambda\hat{i} + \hat{j} - 3\hat{k}$

Area of the triangle $= \frac{\sqrt{195}}{2}$

$\Rightarrow \frac{1}{2}|\vec{a} \times \vec{b}| = \frac{1}{2}\sqrt{195} \Rightarrow |\vec{a} \times \vec{b}|^2 = 195$ ...(i)

Now,

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & \lambda \\ \lambda & 1 & -3 \end{vmatrix}$$

$= \hat{i}(3 - \lambda) - \hat{j}\left(-9 - \lambda^2\right) + \hat{k}(3 + \lambda)$

$|\vec{a} \times \vec{b}| = \sqrt{(3 - \lambda)^2 + (9 + \lambda^2)^2 + (3 + \lambda)^2}$

$\Rightarrow |\vec{a} \times \vec{b}|^2 = (3 - \lambda)^2 + \left(9 + \lambda^2\right)^2 + (3 + \lambda)^2$ ...(ii)

From eqns. (i) and (ii), we get

$(3 - \lambda)^2 + \left(9 + \lambda^2\right)^2 + (3 + \lambda)^2 = 195$

$\Rightarrow 9 + \lambda^2 - 6\lambda + 81 + \lambda^4 + 18\lambda^2 + 9 + \lambda^2 + 6\lambda = 195$

$\Rightarrow \lambda^4 + 20\lambda^2 + 99 = 195$

$\Rightarrow \lambda^4 + 20\lambda^2 - 96 = 0$

$\Rightarrow \left(\lambda^2 + 24\right)\left(\lambda^2 - 4\right) = 0$

$\Rightarrow \lambda^2 - 4 = 0$ or $\lambda^2 + 24 = 0$

$\Rightarrow \lambda = \pm 2 \Rightarrow \lambda^2 = -24$ (not possible)

$\therefore \lambda = \pm 2.$

**Q.58 Answer:**

$\frac{-6k}{(x+2y)^3}$

**Solution:**

Given $x^2 + xy + y^2 = k$

**Differentiating the above function,**

$2x + y + x\frac{dy}{dx} + 2y\frac{dy}{dx} = 0$

$\Rightarrow \frac{dy}{dx} = -\frac{(y+2x)}{x+2y}$

**Differentiating the above function,**

$\frac{d^2y}{dx^2} = \frac{-(x+2y)\left(\frac{dy}{dx}+2\right) + \left(1+2\frac{dy}{dx}\right)(y+2x)}{(x+2y)^2}$

$\Rightarrow \frac{d^2y}{dx^2} = \frac{-3y + \frac{(-3x)(y+2x)}{(x+2y)}}{(x+2y)^2}$

$\Rightarrow \frac{d^2y}{dx^2} = \frac{-3xy-6y^2-3xy-6x^2}{(x+2y)^3}$

$\Rightarrow \frac{d^2y}{dx^2} = \frac{-6(x^2+y^2+xy)}{(x+2y)^3}$

$\Rightarrow \frac{d^2y}{dx^2} = \frac{-6k}{(x+2y)^3}$

**Q.59 Answer:**

$x - y - 1 = 0$

**Solution:**

**We have** $|z-(1+i)|^2 = 2$

$\Rightarrow (x-1)^2 + (y-1)^2 = 2$ (Put $z = x+iy$)

$\Rightarrow x^2 + y^2 = 2(x+y) \ldots (1)$

**Let** $\omega = h + ik = \frac{2}{z} = \frac{2}{x+iy} = \frac{2(x-iy)}{x^2+y^2}$**, so**

$h = \frac{2x}{x^2+y^2}, k = \frac{-2y}{x^2+y^2}$

$\Rightarrow h - k = \frac{2(x+y)}{x^2+y^2} = 1$ **(from equation (1))**

$\therefore$ **Locus of the point** $\omega(h,k)$ **will be** $x - y = 1$

**Q.60 Answer:**

6

**Solution:**

$5a - b + a + 2b = 2(2a+b)$

$2a = b$

$(b+1)^2(a-1)^2 = (ab+1)^2$

$(2a+1)^2(a-1)^2 = \left(2a^2+1\right)^2$

$\left(2a^2-a-1\right)^2 = \left(2a^2+1\right)^2$

$2a^2 - a - 1 = \pm\left(2a^2+1\right)$

**(1)** $2a^2 - a - 1 = 2a^2 + 1 \Rightarrow a = -2$

$(2)\ 2a^2 - a - 1 = -2a^2 - 1$

$4a^2 - a = 0$

$a(4a-1) = 0$

$4a = 1 \Rightarrow a = \frac{1}{4} \Rightarrow b = \frac{1}{2}$

$\frac{1}{a} + \frac{1}{b} = 4 + 2 = 6$

**Q.61 Answer:**

$^{14}C_3$ 3! 15!

**Solution:**

$x_1\ W_2 x_2\ W\ x_3\ W\ x_4$

$x_2, x_3 \geq 2, x_1, x_4 \geq 0$

$x_1 + x_2 + x_3 + x_4 = 15$

**Total integral solutions of** $(x_1, x_2, x_3, x_4)$

$= {}^{14}C_3$

**Total arrangements** $= {}^{14}C_3 \lfloor \underline{3} \mid 15$

**Q.62 Answer:**

57

**Solution:**

$I_1 : 4x + 3y = 3$

$f(x) = y = \frac{3-4x}{3}$ **...(1)**

**since** $f(x)$ **and** $f^{-1}(x)$ **are the mirror images of each other in the line** $y = x$ **hence we find** $f^{-1}(x)$**. now** $y = f(x) \Rightarrow f^{-1}(y) = x$

**from (1)** $x = \frac{3(1-y)}{4}; f^{-1}(y) = \frac{3(1-y)}{4}$

$\therefore f^{-1}(x) = \dfrac{3(1-x)}{4}$

$4y = 3 - 3xL_1 = 3x + 4y - 3 = 0, m_1 = -3/4$

**|||lly** $L_2 = y = -8x$ **with** $m_2 = -8$

**if** $\theta$ **is the acute angle between the lines**

$$\tan\theta = \left|\frac{m_2 - m_1}{1 + m_1 m_2}\right| = \left|\frac{-8 + \frac{3}{4}}{1 + (-8)\left(-\frac{3}{4}\right)}\right| = \left|\frac{-29}{28}\right| \Rightarrow \frac{29}{28}$$

$\Rightarrow a = 29$ and $b = 28$

$\therefore a + b = 29 + 28 = 57$

**Q.63 Answer:**

$nf(x)$

**Solution:**

$$A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$$

**hence det.** $A = \sec^2 x$

$\therefore \det A^\top = \sec^2 x$

**now,** $f(x) = \text{det.}\left(A^\top A^{-1}\right)$

$= \left(\det . A^\top\right)\left(\det . A^{-1}\right)$

$= \left(\det . A^\top\right)(\det . A)^{-1}$

$= \frac{\det \cdot (\mathrm{A}^{\mathrm{T}})}{\det \cdot (\mathrm{A})} = 1$

**hence** $f(x) = 1$

**Q.64 Answer:**

2

**Solution:**

**Let** $\theta = \left(\frac{90}{17}\right)^\circ \Rightarrow 17\theta = 90^\circ$

**Now,** $32 \sin\frac{90^\circ}{17} \cos\frac{180^\circ}{17} \cos\frac{360^\circ}{17} \cos\frac{720^\circ}{17} = 32 \sin\theta \cos 2\theta \cos 4\theta \cos 8\theta = \frac{32 \cos\theta \sin\theta \cos 2\theta \cos 4\theta \cos 8\theta}{\cos\theta} = 2 \times \frac{\sin 16\theta}{\cos\theta}$

$= \frac{2\sin(17\theta - \theta)}{\cos\theta} = \frac{2\sin(90^\circ - \theta)}{\cos\theta} = 2$

**Q.65 Answer:**

$R$ is an equivalence relation

**Solution:**

**Reflexivity For reflexive** $(x, x) \in R$.

$x + 2x = 3x$

which is divisible by 3.

$\Rightarrow$ xRx $\in R$ Hence, xRy is reflexive.

Symmetric Let $x+2y=3\lambda$

$$\Rightarrow \quad x=3\lambda-2y$$

$$\text{Now, } \ y+2x = y+2(3\lambda-2y)$$

$$= y+6\lambda-4y$$

$$= 6\lambda-3y$$

$$\Rightarrow \quad y+2x=3(2\lambda-y)$$

which is divisible by 3 .

$\therefore (x, y) \in R \Rightarrow (y, x) \in R$

Hence, xRy is symmetric.

$$\text{Transifive} \quad x+2y=3\lambda$$

$$\text{and} \quad y+2z=3\mu$$

$$x+3y+2z=3x+3\mu$$

$$\Rightarrow \quad x+2z=3(\lambda+\mu-y)$$

which is divisible by 3.

$\because (x, y) \in$ R and $(y, z) \in R$

$$\Rightarrow \quad (x, z) \in R$$

Hence, xRy is transitive

$\because$ R is reflexive, symmetric and transitive.

$\therefore$ R is an equivalence relation.

**Q.66 Answer:**

$(5,-8,-4)$

**Solution:**

Coordinates of any point $Q$ on the given line are

$(2r+1,-3r-1,8r-10).$

So the direction ratios of $PQ$ are $2r,-3r-1,8r-10.$

Now $PQ$ is perpendicular to the given line.

$$\text{if} \quad 2(2r)-3(-3r-1)+8(8r-10)=0$$

$$\Rightarrow \quad 77r-77=0 \Rightarrow r=1$$

and the coordinates of $Q$, the foot of the perpendicular from $P$ on the line are $(3,-4,-2)$.

Let $R(a, b, c)$ be the reflection of $P$ in the given lines then $Q$ is the mid-point of $PR$

$$\Rightarrow \quad \frac{a+1}{2}=3, \frac{b}{2}=-4, \frac{c}{2}=-2$$

$$\Rightarrow \quad a=5, b=-8, c=-4$$

and the coordinate of the required point are $(5,-8,-4)$.

**Q.67 Answer:**

$\left(\frac{6x+5}{x+1}\right) \cdot e^{x^2}$

**Solution:**

$$f'(x)-\frac{2x(x+1)}{x+1}f(x)=\frac{e^{x^2}}{(x+1)^2}$$

$$\text{I. F. }=e^{\int -2xdx}=e^{-x^2}$$

$$\therefore f(x)\cdot e^{-x^2}=\int\frac{dx}{(x+1)^2}$$

$$\Rightarrow f(x)\cdot e^{-x^2}=-\frac{1}{x+1}+C$$

$$\text{at } x=0, f(0)=5\Rightarrow C=6$$

$$\therefore f(x)=\left(\frac{6x+5}{x+1}\right)\cdot e^{x^2}$$

**Q.68 Answer:**

$\frac{1}{324}$

**Solution:**

**' $a$ ' can take only one value i.e.. 2**

**'b' can take only two values**

**' $c$ ' can take two values**

**'d' can take only one value**

**Hence total favorable ways** $=1.2.2$

$P(E)=\frac{4}{6^4}=\frac{1}{324}$

**Q.69 Answer:**

2

**Solution:**

$$\alpha=\frac{2}{3};\beta=\frac{1}{3}$$

$$x^2-x+\frac{2}{9}=0$$

$$\Rightarrow 9x^2-9x+2=0$$

$$a+b+c=2$$

**Q.70 Answer:**

$\frac{1}{\sqrt{3}}$

**Solution:**

$$\frac{a}{b-c}=\frac{b}{(c-a)}=\frac{c}{(a-b)}=\lambda$$

$$a=\lambda(b-c)\ldots\ldots\ldots(1)$$

$$b=\lambda(c-a)\ldots\ldots\ldots(2)$$

$$c=\lambda(a-b)\ldots\ldots\ldots(3)$$

$$(1)+(2)+(3)\Rightarrow a+b+c=0$$

$$\frac{|\bar{\alpha}|}{|\bar{\beta}|}=\frac{\sqrt{a^2+b^2+c^2}}{\sqrt{2\left(a^2+b^2+c^2\right)}-2(ab+bc+ca)}$$

**By squaring** $\mathrm{a}+\mathrm{b}+\mathrm{c}=0$

$$\Rightarrow\left(a^2+b^2+c^2\right)=-2(ab+bc+ca)$$

$$\frac{|\bar{\alpha}|}{|\bar{\beta}|}=\frac{1}{\sqrt{3}}$$

**Q.71 Answer:**

—

**Solution:**

$3^{2n+2}-8n-9=9^{n+1}-8(n+1)-1=(8+1)^{n+1}-8(n+1)-1$
$=8^{n+1}+\left(\begin{array}{c} n+1 \\ 1 \end{array}\right)8^n+\ldots+\left(\begin{array}{c} n+1 \\ n-1 \end{array}\right)8^2$, **which is divisible by** $8^2=2^6$.

**Q.72 Answer:**

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**Solution:**

**We have** $\left(1+\frac{1}{\mathrm{n}}\right)^{\mathrm{n}+\mathrm{x}_{\mathrm{n}}}=\mathrm{e}$ **...(1)**
**Taking log on both sides of equation (1), we get**

$$\Rightarrow (n+x_n)\ln\left(1+\frac{1}{n}\right)=1$$
$$\Rightarrow n+x_n=\frac{1}{\ln\left(1+\frac{1}{n}\right)}$$
$$\Rightarrow x_n=\frac{1}{\ln\left(1+\frac{1}{n}\right)}-n\ldots(1)$$

**Let** $\frac{\mathrm{n}+1}{\mathrm{n}}=\mathrm{u}$
$\Rightarrow nu=n+1\Rightarrow n=\frac{1}{u-1}$
$\therefore \mathrm{Lim}_{n\to\infty}\,x_n=\mathrm{Lim}_{u\to1}\left(\frac{1}{\ln u}-\frac{1}{u-1}\right)=\mathrm{Lim}_{u\to1}\,\frac{(u-1)-\ln u}{(u-1)\ln u}\left(\frac{0}{0}\right)$ **form** $=\mathrm{Lim}_{u\to1}\,\frac{1-\frac{1}{u}}{\frac{u-1}{u}+\ln u}=\mathrm{Lim}_{u\to1}\,\frac{\frac{1}{u^2}}{\frac{1}{u^2}+\frac{1}{u}}=\frac{1}{2}$

**Q.73 Answer:**

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**Solution:**

$$\int_0^{100}\{\sqrt{x}\}dx=\int_0^{100}\{\sqrt{x}-[\sqrt{x}]\}dx=\int_0^{100}\sqrt{x}dx-\int_0^{100}[\sqrt{x}]dx$$
$$=\frac{2}{3}x^{3/2}\Big|_0^{100}-\int_0^1 0dx+\int_1^4 1dx+\int_4^a 2dx+\ldots.+\int_{19^2}^{10^2}9dx$$
$$=\frac{2}{3}\left[10^3\right]-\left[0+1\left(2^2-1^2\right)+2\left(3^2-2^2\right)+2\left(3^2-2^2\right)+3\left(4^2-3^2\right)+\ldots..+9\left(10^2-9^2\right)\right]$$
$$=\frac{2}{3}10^2=9\times10^2+\left(1^2+2^2+\ldots.+9^2\right)=\frac{2}{3}\times10^3-900+\frac{9(9+1)(18+1)}{6}=\frac{2000}{3}-900+\frac{9\times10\times19}{6}$$
$$=\frac{2000}{3}\left(900-\frac{1710}{6}\right)=\frac{2000}{3}-\frac{3690}{6}=\frac{310}{6}.$$

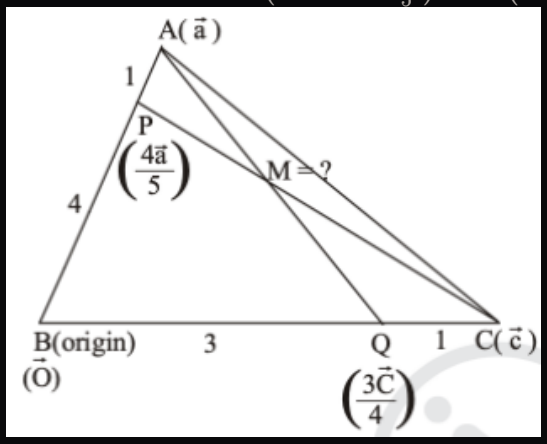
**Q.74 Answer:**

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**Solution:**

Vector equation of $AQ$ is $\vec{r}_1=\vec{a}+\lambda\left(\vec{a}-\frac{3\vec{c}}{4}\right)$
**and vector equation of** $CP$ **is** $\vec{r}_2=\overrightarrow{c}+\mu\left(\overrightarrow{c}-\frac{4\overrightarrow{a}}{5}\right)$

Hence $\vec{r}_1 = \vec{r}_2$ gives, $\left(1+\lambda+\frac{4\mu}{5}\right)\vec{a} = \left(1+\mu+\frac{3\lambda}{4}\right)\vec{c}$




$\therefore 1+\lambda+\frac{4\mu}{5}=0$ and $1+\mu+\frac{3\lambda}{4}=0$

Solving, $\lambda = -\frac{1}{2}$ and

$\mu = -\frac{5}{8}$

$\therefore$ p.v. of $M = \vec{a} - \frac{1}{2}\vec{a} + \frac{3\vec{c}}{8} = \frac{4\vec{a}+3\vec{c}}{8}$

$\overrightarrow{MC} = \vec{c} - \frac{4\vec{a}+3\vec{c}}{8} = \frac{5\vec{c}-4\vec{a}}{8}$

And $\overrightarrow{PC} = \overrightarrow{c} - \frac{4\overrightarrow{a}}{5} = \frac{5\overrightarrow{c}-4\overrightarrow{a}}{5}$

$\therefore \frac{MC}{PC} = \frac{5}{8}$

$\Rightarrow a = 5$ and $b = 8$;

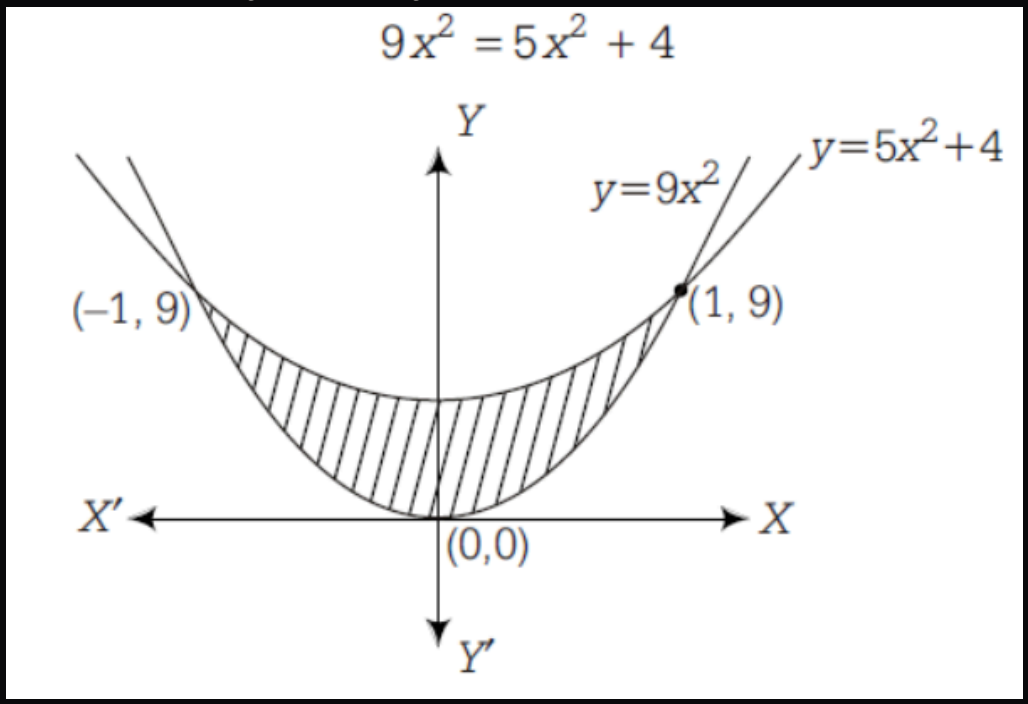
$\therefore a + b = 13$

**Q.75 Answer:**

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**Solution:**

Intersecting point of $y = 9x^2$ and $y = 5x^2 + 4$ is given by



$$\Rightarrow \quad 4x^2 = 4$$
$$\Rightarrow \quad x^2 = 1$$
$$\Rightarrow \quad x = \pm 1$$
$$\therefore \quad y = 9(\pm 1)^2 = 9$$

**Hence, intersecting points are** $(\pm 1, 9)$.

$$\therefore \text{ Required area } = 2\int_0^1 \left[\left(5x^2+4\right) - \left(9x^2\right)\right] dx$$
$$= 2\int_0^1 \left(4 - 4x^2\right) dx$$
$$= 2\left[4x - \frac{4x^3}{3}\right]_0^1$$
$$= 2\left(4 - \frac{4}{3}\right) - (0-0) = 2 \times \frac{8}{3}$$
$$= \frac{16}{3} \text{ sq units}$$